GEOMETRY BY CONSTRUCTION: A GEOMETRIC PRIOR BOTTLENECK

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Paper under double-blind review

ABSTRACT

The conditional entropy bottleneck (CEB) compresses representations against a label-conditional reference whose geometry is left entirely to the optimizer. We show that this reference has two failure channels: in deterministic settings the CEB knob is dead (a known Caveat), and, more fundamentally, the certificate is blind to the relationships between labels: no cross-class term constrains the geometry among the class priors. We close this channel by declaring the geometry: the prior is confined to a qualified geometric family computed statelessly at every training step: an orthogonal class frame (OPB) for classification and an isometric regression axis (EPB) for regression, both with prediction rules tied to the geometry by construction (the zero-parameter anchor-energy classifier and the tied projection head). The framework preserves the CEB certificate for any stateless conditional prior and, under an explicit realizability assumption, yields β-controlled anchor-alignment bounds that imply separation transfer; without realizability, separation remains conditional on the observed anchor mismatch. The construction removes the unspecified inter-class geometry of the reference. Across nine datasets and four backbones, GPB is best or tied-best in 7 of 12 settings and dominates CEB everywhere, and the anchor scale converts the β-knob from a collapse dial into a plateau dial.

1 INTRODUCTION

The information bottleneck (IB) principle (Tishby et al., 2000) seeks a representation Z of an input X that retains the information needed to predict the label Y while discarding the rest. The variational IB (VIB) (Alemi et al., 2017) bounds the compression term with a fixed marginal prior $\mathcal{N}(0, I)$; the conditional entropy bottleneck (CEB) (Fischer, 2020) sharpens the bound to the residual information by replacing the marginal prior with a trainable label-conditional reference $b(z|y)$, certified by the gap $\mathrm{ReX} = \mathbb{E}[\mathrm{KL}(q(z|x) \,\|\, b(z|y))] \geq I(X; Z|Y)$. We call this gap the *certificate*; its failure modes are our starting point.

The problem: a reference with no declared geometry. Two facts about CEB are established. First, *the knob is dead*: in the deterministic setting $Y = f(X)$, every compression weight selects the same corner of the information plane (Kolchinsky & Tracey, 2019); this is settled background, repaired by their squared compression term, and we return to it only as a mechanism connection (Proposition A.4, Appendix). Second, and less recognized, *the certificate is blind to the relationships between labels*: the CEB compression term is a sum of per-class matching terms (each sample's KL touches only its own class prior, and no term couples two distinct class priors), so at its class-wise matched optimum it reduces to $I(X; Z|Y)$, irrespective of the inter-class geometry realized by those matched conditionals (Proposition 2.1). Class means may merge, coincide, or point anywhere without changing the certificate; on the family $T_\alpha = B_\alpha \cdot Y$ that interpolates from collapse to the exact copy of the label, the certificate reads zero across the whole retention axis (Proposition A.2, Appendix). CEB therefore compresses against a reference whose inter-class geometry the optimizer may choose at will: what a properly compressed representation should look like (class means separated, ordered, or laid out on a frame) is exactly what the certificate leaves unspecified.

A geometric prior bottleneck. We close this gap by *declaring* the geometry of the reference. The prior $p(z|y)$ is confined to a *qualified geometric family*: label-conditional Gaussians with a bounded diagonal covariance and a mean map that encodes the label structure at a fixed, declared scale: a two-sided separation condition for discrete labels and a two-sided metric condition for continuous labels (Section 2.1). The prior is recomputed statelessly at every step, so the degenerate references — merged classes, a collapsed point, an arbitrarily rescaled geometry — are excluded by construction rather than by the optimizer's good behavior. We call this family of objectives a *geometric prior bottleneck* (GPB). Two instantiations realize the principle: *OPB* (orthogonal prior bottleneck) for classification, where the prior means form the scaled orthogonal frame $a\, q_{\theta,k}$ (every pair of distinct classes at distance $\sqrt{2}\, a$, with class-mean Gram matrix $a^2 I_K$ by construction), and *EPB* (equidistant prior bottleneck) for regression, where the prior means lie on a unit axis, $\rho\, \tilde{y}\, u$, so latent distances equal ρ times standardized label distances exactly. Both keep the CEB posterior, a single KL, and the residual certificate unchanged; the classification head is the anchor-energy classifier $\propto \exp(-\frac{1}{2\tau^2}\|z - a\, q_{\theta,k}\|^2)$ and the regression head the tied projection $\hat{\tilde{y}} = u^\top z/\rho$: prediction rules tied to the prior geometry by construction, with no free parameters. The constraints bind the *reference*, not the deployed representation; the framework constrains the geometric *form* of the label code and does not claim that the certificate meters retained label information.

Conditional theoretical guarantees and scope. The theory has two levels. The certificate gap identity holds pointwise for any stateless prior (Proposition 2.3), so CEB's residual bound transfers without a realizability assumption. The stronger $O(1/\beta)$ anchor-mismatch bound, $\mathbb{E}[D(Y)] \leq (C_{\text{align}}+\varepsilon)/\beta$, is an idealized comparison-solution result: it requires deterministic labels, an encoder and posterior family that can realize $q(z|x) = p_\theta(z|y)$, and a compatible prediction head. Under these assumptions, $C_{\text{align}} \leq \ln K$ for deterministic K-class classification and $C_{\text{align}} = \tau^2/\rho^2$ for the isometric regression axis (Theorem 2.1, parts (a) and (b)). We do not assume these premises for every benchmark task. When the aligned comparison solution is infeasible, only the mismatch-dependent separation inequality remains available; the experiments then measure the realized mismatch and geometry directly. Under the stated assumptions, a triangle-inequality transfer gives posterior separation up to terms of order $1/\sqrt{\beta}$ (Proposition 2.4 and its corollaries). The framework pins exactly what CEB left free: the geometry of the reference.

Experiments. Across nine datasets and four backbones, GPB is best or tied-best in 7 of 12 settings and dominates CEB in all 12; the mechanism-level validation confirms the story end to end: the constructed geometry transfers to the posterior (nearly orthogonal class centers on classification, an on-axis posterior on regression), and prediction survives with the zero-parameter geometric heads, matching or exceeding the free classifier. A separate exact-realizability synthetic control shows rapid β-dependent mismatch decay without loss of prediction accuracy, while making explicit that this controlled check does not extend the theorem to the real-data benchmarks.

Our contributions are: (1) the identification and formalization of the missing cross-label geometric constraint in CEB (Proposition 2.1); (2) the GPB framework and its two instances, OPB and EPB, with prediction rules tied to the prior geometry; (3) a conditional alignment-to-separation analysis under an explicit realizability assumption, together with unconditional certificate preservation and mismatch-dependent separation bounds for any qualified geometric prior, instantiated by the orthogonal frame and the isometric axis; and (4) mechanism-level empirical evidence that the constructed geometry is transferred to the posterior and is sufficient for prediction; and (5) a controlled exact-realizability validation of the conditional alignment scaling, explicitly separated from the real-data benchmark claims.

2 Method

2.1 From CEB to GPB

Let X be the input and Y the label with K classes; the encoder induces the Markov chain $Y - X - Z$. Following Fischer (2020), the variational CEB objective is $\min_{e,b,c} \langle \log e(z|x)\rangle - \langle \log b(z|y)\rangle - \gamma \langle \log c(y|z)\rangle$ over the stochastic encoder e, the backward encoder b, and the classifier c. We use the diagonal Gaussian posterior $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \operatorname{diag}(\sigma_\phi^2(x)))$, the loss convention CE $+\beta$ KL ($\beta = 1/\gamma$), and the deterministic code $z = \mu_\phi(x)$ at evaluation. The central object is the residual certificate: for any conditional prior $r(z|y)$,

$$\mathbb{E}\big[\mathrm{KL}(q_\phi(z|x) \,\|\, r(z|y))\big] = I(X;Z|Y) + \mathbb{E}_y\big[\mathrm{KL}(q_\phi(z|y) \,\|\, r(z|y))\big] \;\geq\; I(X;Z|Y), \qquad (1)$$

by Fischer's chain-rule identity. CEB instantiates $r(z|y) = b_\theta(z|y)$; VIB fixes $r = \mathcal{N}(0, I)$. The certificate's failure mode is that its compression term carries no cross-class structure:

Proposition 2.1 (The certificate records nothing about inter-label geometry)**.** *The CEB compression term contains no explicit term coupling two distinct class priors: it is a sum of per-class matching terms, each sample's KL touching only its own class prior. Consequently, at its class-wise matched optimum ($b = q(z|y)$), the certificate reduces to*

$$\mathrm{ReX} = I(X; Z|Y), \tag{2}$$

irrespective of the inter-class geometry realized by the matched conditionals: class means merged, coincident, or placed in arbitrary directions all yield the same certificate value. (The prediction term couples classes through the softmax; the statement is at a fixed encoder with a matched backward encoder.)

At the matched optimum the certificate thus carries no information about the relationships among the class priors, and on the retention family $T_\alpha = B_\alpha \cdot Y$ of Kolchinsky & Tracey (2019) it reads zero from total collapse to the full copy (Proposition A.2, Appendix). We close this channel by *declaring* the geometry of the reference.

Let $\mathcal{G}$ be a *qualified geometric prior family*: a set of label-conditional distributions $p_\theta(z|y)$ that depend on y alone, computed from the prior encoder, with a bounded diagonal covariance,

$$p_\theta(z|y) = \mathcal{N}\big(m_\theta(y),\, \Sigma_\theta(y)\big), \qquad \Sigma_\theta(y) \text{ diagonal}, \qquad \Sigma_\theta(y) \;\preceq\; \tau^2_{\max} I_d \text{ for all } y, \tag{3}$$

and a mean map that encodes the label structure at a fixed, declared scale: for discrete labels a two-sided separation condition,

$$\Delta \;\le\; \|m_\theta(y) - m_\theta(y')\| \;\le\; \bar{\Delta} \text{ for all } y \ne y', \tag{4}$$

and for continuous labels with a declared label metric d_Y,

$$\rho_{\min}\, d_Y(y, y') \;\le\; \|m_\theta(y) - m_\theta(y')\| \;\le\; \rho_{\max}\, d_Y(y, y'). \tag{5}$$

The declared constants $\Delta, \bar{\Delta}, \rho_{\min}, \rho_{\max}$ are fixed before training and not trained. The lower bounds encode the label distinctions; the upper bounds close the scale escape: without them the family would contain $m_\theta(y) = s\, m_0(y)$ for arbitrarily large s, and the objective might lack a finite optimum, making "geometry by construction" vacuous. The GPB objective is

$$\mathcal{L}_{\mathrm{GPB}} = \mathbb{E}_{(x,y)}\Big[\mathrm{CE}\big(c_\psi(z), y\big) + \beta\ \mathrm{KL}\big(q_\phi(z|x) \,\|\, p_\theta(z|y)\big)\Big], \qquad z \sim q_\phi(z|x), \qquad p_\theta \in \mathcal{G}. \tag{6}$$

2.2 Geometric instantiations

2.2.1 OPB: an orthogonal class frame (classification)

For classification, let $K \le d$. The prior encoder g_θ is a single unactivated linear map, evaluated on the all-class one-hot matrix E_K at every forward pass, giving the raw class-mean matrix

$$U_\theta = g_\theta(E_K)^\top \in \mathbb{R}^{d \times K}, \tag{7}$$

orthonormalized by the thin polar factor

$$Q_\theta = U_\theta\big(U_\theta^\top U_\theta\big)^{-1/2}, \qquad Q_\theta^\top Q_\theta = I_K, \tag{8}$$

computed via the thin SVD $U_\theta = W \Sigma V^\top$, $Q_\theta = W V^\top$ (no sign convention needed). The polar factor is exactly equivariant under relabeling of the classes, $Q_\theta(U_\theta P) = Q_\theta(U_\theta)\, P$ (derivation in Appendix A.3), so no class index plays a distinguished role. We make one standing assumption, verified, not imposed, in the implementation:

Assumption 2.1 (Full rank)**.** *U_θ has full column rank at every training step, so that Eq. equation 8 and its backward pass are defined. The condition holds almost surely at the identity initialization, and $\sigma_{\min}(U_\theta)$ is monitored along training (Appendix A.3).*

The geometric prior replaces the raw means by the scaled frame

$$p_\theta^{\mathrm{OPB}}(z|y = k) = \mathcal{N}\big(a\, q_{\theta,k},\, \tau^2 I_d\big), \qquad q_{\theta,k} := Q_\theta[:, k], \tag{9}$$

with anchor scale $a > 0$ and *fixed shared* prior variance τ^2 (default 1.0): a learnable per-class variance would give the reference an extra channel to absorb the compression cost by inflating its covariance, precisely the degeneracy the construction excludes. Since $Q_\theta^\top Q_\theta = I_K$, the class-mean Gram matrix is $a^2 I_K$ by construction:

$$a_i^\top a_j = a^2 \delta_{ij}, \qquad \|a_k\| = a, \qquad \|a_i - a_j\|^2 = 2a^2 \ (i \neq j). \tag{10}$$

The frame is stateless: an immediate function of the current prior-encoder parameters, computed from the all-class table: no EMA, no prototypes, no batch statistics. The single-KL objective is

$$\mathcal{L}_{\mathrm{OPB}} = \mathbb{E}_{(x,y)}\Big[\mathrm{CE}\big(c_\psi(z), y\big) + \beta\ \mathrm{KL}\big(q_\phi(z|x) \,\|\, p_\theta^{\mathrm{OPB}}(z|y)\big)\Big], \qquad z \sim q_\phi(z|x). \tag{11}$$

The main classifier is the anchor-energy classifier, whose scale is fixed by construction:

Proposition 2.2 (Anchor-energy classifier equivalence)**.** *The scores*

$$s_k(z) = -\frac{\|z - a\, q_{\theta,k}\|^2}{2\tau^2} + \log \pi_k \tag{12}$$

are the class-k Gaussian log posterior up to a class-independent term; for uniform classes the equivalent linear classifier has weights $w_k = (a/\tau^2)\, q_{\theta,k}$: equal norms and pairwise orthogonal rows. The prediction rule therefore uses exactly the same geometry as the prior, with no free classifier scale (proof in Appendix A.3).

The posterior logvar head is zero-initialized and the prior encoder identity-initialized ($U_\theta = [e_1, \ldots, e_K]$, full column rank), so training starts at $\mathrm{KL} \approx a^2/(2\tau^2)$ with a stable polar backward. The plain linear classifier is kept in the experiments as a free-head variant; it does not enjoy the fixed-scale condition of Proposition A.4.

2.2.2 EPB: AN EQUIDISTANT REGRESSION AXIS (REGRESSION)

For a one-dimensional continuous label, standardize once on the training set, $\tilde{y} = (y - y_{\mathrm{center}})/y_{\mathrm{scale}}$, with the statistics frozen for validation and test. Let $W \in \mathbb{R}^{d\times 1}$ be a trainable direction and $u = W/\|W\|$ its per-step normalization. The isometric prior is

$$p^{\mathrm{EPB}}(z|y) = \mathcal{N}\big(\rho\, \tilde{y}\, u,\ \tau^2 I_d\big), \qquad \|\mu_p(y_i) - \mu_p(y_j)\| = \rho\, |\tilde{y}_i - \tilde{y}_j|, \tag{13}$$

with isometric scale $\rho > 0$ and fixed shared prior variance τ^2: latent distances equal ρ times standardized label distances exactly, at every scale. Order and magnitude of label differences are encoded with equality, which qualifies the family (Eq. equation 3 with $\tau^2_{\max} = \tau^2$ and Eq. equation 5 with $\rho_{\min} = \rho_{\max} = \rho$ on the standardized label metric). The main variant uses the trainable normalized axis and no bias (the fixed-axis and unconstrained-scale ablations are in Appendix A.3). The prediction head is the *tied projection* $\hat{\tilde{y}} = u^\top z/\rho$, the regression analogue of the anchor-energy classifier: geometrically consistent, with no free direction or scale that could bypass the prior; the ordinary regression head is kept as a free-head variant. The objective is $L_{\mathrm{reg}}(\hat{y}, y) + \beta\ \mathrm{KL}\big(q_\phi(z|x) \,\|\, p^{\mathrm{EPB}}(z|y)\big)$: a single KL, stateless.

2.3 CONDITIONAL THEORETICAL ANALYSIS

The theory separates an unconditional certificate statement from a conditional alignment statement. The gap identity below holds for every stateless conditional prior. The $O(1/\beta)$ alignment bound requires a feasible aligned comparison solution and is therefore an idealized realizability result, not a universal guarantee for arbitrary datasets or finite encoder classes. In particular, $Y = f(X)$ alone does not imply that a given encoder can recover Y or represent the prior covariance; the comparison solution must be feasible in the chosen model and head families. When it is not feasible, the separation inequalities below remain valid in terms of the realized mismatch D, but no $O(1/\beta)$ upper bound on that mismatch is claimed.

The first, unconditional link holds pointwise:

Proposition 2.3 (A stateless geometric prior preserves the CEB certificate)**.** *For any prior family $\mathcal{G}$ whose members are conditional priors independent of x, and for every parameter value θ at which $p_\theta \in \mathcal{G}$ is defined, the gap identity of Eq. equation 1 applies pointwise in θ:*

$$\mathbb{E}\Big[\mathrm{KL}\big(q_\phi(z|x) \,\|\, p_\theta(z|y)\big)\Big] = I_{q_\phi}(X; Z|Y) + \mathbb{E}_Y\Big[\mathrm{KL}\big(q_\phi(z|Y) \,\|\, p_\theta(z|Y)\big)\Big] \ \geq\ I_{q_\phi}(X; Z|Y). \tag{14}$$

A batch-dependent prior would break the conditional-prior argument.

Proposition 2.4 (Mismatch-dependent separation transfer for any qualified geometric prior)**.** *Let $\mathcal{G}$ be qualified with variance bound $\tau^2_{\max}$.* (a) Discrete labels. *With posterior class centers $m_k := \mathbb{E}[\mu_\phi(X)|Y=k]$ and class-wise anchor mismatch*

$$D_k := \frac{1}{2\tau^2_{\max}}\,\mathbb{E}\Big[\|\mu_\phi(X) - m_\theta(k)\|^2 \;\Big|\; Y=k\Big], \tag{15}$$

for every pair $i \neq j$,

$$\|m_i - m_j\| \;\geq\; \|m_\theta(i) - m_\theta(j)\| - \sqrt{2\tau^2_{\max}D_i} - \sqrt{2\tau^2_{\max}D_j} \;\geq\; \Delta - \sqrt{2\tau^2_{\max}D_i} - \sqrt{2\tau^2_{\max}D_j}, \tag{16}$$

using Eq. equation 4. (b) Continuous labels. *For Borel-measurable X, Y with integrable $\mu_\phi(X)$, the regular conditional posterior center $m(y) := \mathbb{E}[\mu_\phi(X) \mid Y=y]$ and pointwise mismatch*

$$D(y) := \frac{1}{2\tau^2_{\max}}\,\mathbb{E}\Big[\|\mu_\phi(X) - m_\theta(y)\|^2 \;\Big|\; Y=y\Big] \tag{17}$$

are defined for P_Y-a.e. y, and for P_Y-a.e. pair $y \neq y'$,

$$\|m(y) - m(y')\| \;\geq\; \|m_\theta(y) - m_\theta(y')\| - \sqrt{2\tau^2_{\max}D(y)} - \sqrt{2\tau^2_{\max}D(y')} \;\geq\; \rho_{\min}\, d_Y(y,y') - \sqrt{2\tau^2_{\max}D(y)} - \sqrt{2\tau^2_{\max}D(y')}. \tag{18}$$

Under the realizability conditions stated explicitly below, the objective controls the mismatch through a comparison-solution argument, in one theorem stated for the two label-space structures:

Theorem 2.1 (Conditional anchor alignment at near-optima under realizability)**.** (a) Deterministic K-class classification. *Assume $Y = f(X)$, a cross-entropy prediction loss, a qualified family whose diagonal covariances the posterior can represent, and an encoder family containing class-constant maps, so that the aligned solution $q(z|x) = p_\theta(z|y)$ is feasible; assume further that the classifier family contains the Bayes posterior of the aligned mixture,*

$$c_k(z) = \frac{\pi_k\,|\Sigma_\theta(k)|^{-1/2}\, e^{-\frac{1}{2}(z-m_\theta(k))^\top \Sigma_\theta(k)^{-1}(z-m_\theta(k))}}{\sum_j \pi_j\,|\Sigma_\theta(j)|^{-1/2}\, e^{-\frac{1}{2}(z-m_\theta(j))^\top \Sigma_\theta(j)^{-1}(z-m_\theta(j))}}. \tag{19}$$

Then for any solution (ϕ,θ,ψ) with objective value $L \leq L^* + \varepsilon$, where L^* is the infimum of Eq. equation 6, the expected anchor mismatch is controlled by β:

$$\mathbb{E}\big[D(Y)\big] \;\leq\; \frac{C_{\mathrm{G}} + \varepsilon}{\beta} \;\leq\; \frac{\ln K + \varepsilon}{\beta}, \tag{20}$$

where $C_{\mathrm{G}} := H(Y \mid Z_{\mathrm{align}})$ is the cross-entropy of the Bayes classifier of Eq. equation 19 on the aligned mixture. In the shared-covariance orthogonal instance the determinant factors cancel and this Bayes posterior reduces exactly to the anchor-energy classifier of Eq. equation 12.

(b) Deterministic continuous regression. *Assume $Y = f(X)$ with a non-negative regression loss, a qualified family, a posterior able to realize $q(z|x) = p_\theta(z|y)$, and the tied projection head of the isometric instance. For the standardized squared loss, $\hat{\tilde{y}} = u^\top z/\rho$ with $z \sim p_\theta(z|y)$ gives $\hat{\tilde{y}} = \tilde{y} + (\tau/\rho)\,\eta$ with $\eta \sim \mathcal{N}(0,1)$, so the aligned risk is $C_{\mathrm{align}} = \tau^2/\rho^2$. Then for any solution with $L \leq L^* + \varepsilon$,*

$$\mathbb{E}\big[D(Y)\big] \;\leq\; \frac{C_{\mathrm{align}} + \varepsilon}{\beta}, \tag{21}$$

with $D(y)$ as in Eq. equation 17. (The scope of the deterministic comparison-solution argument is discussed in Appendix A.1.)

The theorem should be read as a conditional comparison-solution result. If the aligned posterior is not realizable by the chosen encoder and posterior family, or if the aligned predictor is outside the head class, the displayed theorem does not upper-bound $\mathbb{E}[D(Y)]$; the mismatch-dependent inequalities of Proposition 2.4 are the applicable statements.

Corollary 2.1 (Conditional separation at near-optimal solutions)**.** (a) Classification. *Under Theorem 2.1(a), for classes $i \neq j$ with $\pi_i, \pi_j > 0$,*

$$\|m_i - m_j\| \;\geq\; \|m_\theta(i) - m_\theta(j)\| - \sqrt{\frac{2(C_{\mathrm{G}}+\varepsilon)}{\beta\,\pi_i}} - \sqrt{\frac{2(C_{\mathrm{G}}+\varepsilon)}{\beta\,\pi_j}}. \tag{22}$$

(b) Regression. *Under Theorem 2.1(b), for any $t > 0$, on the label set $S_t := \{y : D(y) \le t\}$ of P_Y-measure at least $1 - (C_{\text{align}} + \varepsilon)/(\beta t)$, the pointwise transfer of Proposition 2.4(b) yields, for all $y, y' \in S_t$,*

$$\|m(y) - m(y')\| \;\ge\; \rho_{\min}\, d_Y(y, y') - 2\sqrt{2\tau_{\max}^2\, t}; \tag{23}$$

for EPB ($\rho_{\min} = \rho$, $\tau_{\max} = \tau$) the bound reads $\rho\, d_Y(y, y') - 2\sqrt{2\tau^2 t}$.

The orthogonal instance upgrades these statements with a constructive fact:

Proposition 2.5 (Positive rate cost of class-independent means)**.** *If the posterior mean is class-independent, $\mathbb{E}[\mu_\phi(X)|Y = k] = c$ for every k, the mean-mismatch part of the rate satisfies*

$$\mathbb{E}\Big[\tfrac{1}{2\tau^2}\|\mu_\phi(X) - a\, q_{\theta,Y}\|^2\Big] \;\ge\; \frac{a^2}{2\tau^2}\Big(1 - \sum_{k=1}^{K} \pi_k^2\Big), \tag{24}$$

which for uniform classes reads $a^2(1 - 1/K)/(2\tau^2)$: the VIB corner $\mu \equiv 0$ is a penalized configuration rather than a free optimum, whenever $a > 0$ and two classes have positive probability.

Corollary 2.2 (The orthogonal instance instantiates the conditional separation transfer)**.** *The orthogonal instance is qualified with $\tau_{\max}^2 = \tau^2$ and $\Delta = \bar{\Delta} = \sqrt{2}\, a$; Proposition 2.4 specializes to*

$$\|m_i - m_j\| \;\ge\; \sqrt{2}\, a - \sqrt{2\tau^2 D_i} - \sqrt{2\tau^2 D_j}, \qquad D_k := \frac{1}{2\tau^2}\,\mathbb{E}\Big[\|\mu_\phi(X) - a\, q_{\theta,k}\|^2 \;\Big|\; Y = k\Big]. \tag{25}$$

Scope. The geometric constraints bind the reference, not the deployed representation: at any frame rotation there exists a full-retention label code with a zero certificate (Remark A.3, Appendix), so the framework constrains the geometric *form* of the label code (precisely what CEB left unspecified), and does not claim that the certificate meters retained label information. The proofs, the dead-knob connection (the anchored reference supplies the quadratic displacement behind the strict convexity of Proposition A.4), and the implementation details are in the appendix.

3 Experiments

The experiments address two questions. First, does replacing CEB's unconstrained reference with a constructed geometry (an orthogonal class frame or an isometric axis) cost or gain anything in end-to-end prediction across heterogeneous settings? Second, does prediction survive when the prediction rule is tied to that geometry by construction (the anchor-energy classifier and the tied projection head), so that the free head's degrees of freedom are removed? The main sweep answers both on twelve dataset–backbone settings; the compression–accuracy and realized-$\mathbb{E}[\mathrm{KL}]$ curves are reported in Section 3.5, adversarial robustness in Section 3.7, the geometric mechanism validation in Section 3.8, and the remaining mechanism-level checks (nearest-anchor sufficiency, memorization) in the supplementary analysis.

3.1 Experimental setup

Datasets. Twelve settings over nine datasets and four backbones, spanning image, graph, and text modalities. *Classification*: MNIST (MLP and CNN backbones); ImageNet-100 (MLP and CNN) on frozen ResNet-50 features (layer-3 activations pooled to 4×4); Cora, transductive node classification (GCN); IMDb sentiment (RNN); AG News (RNN) on frozen BERT layer-10 token features reduced by PCA to 50 dimensions. *Regression*: California Housing (MLP); STS-B (RNN with frozen GloVe-100d embeddings); ZINC-12k molecular graphs (GCN, the official 10k/1k/1k split); AgeDB face age (MLP and CNN). Where no official split exists we use a fixed $60/20/20$ train/validation/test partition (seed 42); on MNIST 10k of the 60k training images are held out for validation. Regression targets are standardized to $[0, 1]$ on the training set (the test metric is re-normalized).

Backbones and protocol. The MLP uses hidden layers 512–256; the CNN two convolutional blocks (channels 32, 64) followed by a 256-dim hidden layer; the GCN matches the MLP sizes; the RNN is a two-layer LSTM with 512 and 256 units (a single 256-unit layer with max time-pooling on STS-B). All methods share the backbone, dropout 0.2, and bottleneck dimension $d = 256$. Every run trains with Adam (learning rate 10^{-3}; 10^{-2} on Cora), at most 100 epochs with early stopping on the validation metric (patience 10), batch size 512 (128 on ZINC), and evaluates the checkpoint with the best validation score on the test set. GPB and GPB-L train five seeds per configuration (seeds 0–4); each variational baseline trains fifteen seeds per β value (seeds 0–14), so that every method trains a matched total budget ($6 \times 3 \times 5 = 6 \times 15 = 90$ runs per method). The tables report the mean over the runs. Classification early-stops on validation macro-AUC and reports test accuracy; regression early-stops on validation R^2 and reports test R^2.

Baselines. The plain backbone (Base); VIB (Alemi et al., 2017) with the fixed $\mathcal{N}(0, I)$ prior; SVIB, the squared compression term of Kolchinsky & Tracey (2019) ($\mathrm{CE} + \beta\ \mathrm{KL}^2$); NIB (Kolchinsky et al., 2019), the noise-injection bottleneck with the pairwise-distance upper bound on $I(X; M)$; CEB (Fischer, 2020) with the trainable label-conditional backward encoder; and the supervised β-DVCCA of Abdelaleem et al. (2025), which augments the VIB loss with an input reconstruction term, $\mathrm{CE} + \beta\,(\mathrm{KL} + \mathrm{MSE}_{\mathrm{recon}})$. All variational methods share the encoder, the posterior heads, and the loss convention of Eq. equation 6.

External baselines. Two non-IB comparison methods complete the reference set. *NCBD* (Koromilas et al., 2026) (Neural Collapse by Design) normalizes features and trainable class prototypes to the unit hypersphere and replaces the classification loss by the NONL contrastive objective, whose global optimum is the simplex ETF; its temperature τ occupies the β slot of the sweep (grid $\{0.05, 0.1, 0.2, 0.3, 0.5, 1\}$), and it applies to classification settings only. *AdaCap* (Belucci et al., 2026) targets small-sample regression: a permutation-based contrastive loss plus a Tikhonov closed-form output layer $\hat{y} = H\,(H^\top H + \lambda I)^{-1} H^\top y$ with learnable λ, initialized from the swept grid $\{10^{-2}, \ldots, 10^{3}\}$; at evaluation the readout is refit on the training-set representations. It applies to regression settings only. Both train under the shared backbones, splits, early-stopping protocol, and five seeds per configuration.

Our variants. *GPB* denotes our method in the tables. On classification rows it is *OPB*, the orthogonal-frame prior of Eq. equation 9 with the anchor-energy classifier of Eq. equation 12: the prediction rule has no free parameters, its means and variances are exactly the prior's, and $K \leq d$ holds in all settings ($\max K = 100$ on ImageNet-100); on regression rows it is *EPB*, the isometric prior of Eq. equation 13 with the tied projection head $\hat{\hat{y}} = u^\top z/\rho$. *GPB-L* is the free-head ablation: the identical prior and KL, with an ordinary trainable linear classifier (or regression head) on z, so prediction is free to bypass the constructed geometry. The frame is orthonormalized from the all-class mean matrix at every forward pass by the thin SVD polar factor of Eq. equation 8 ($Q_\theta = WV^\top$); the prior encoder is identity-initialized (initial anchors $e_1, \ldots, e_K$, initial $\mathrm{KL} \approx a^2/2$ under unit variances), the variance heads are zero-initialized, and the construction is stateless: no EMA, no prototypes, no batch statistics. Six attribution variants (NCM-learn/NCM-ortho, CEB-energy/CEB-tied, OPB-FS/EPB-FS) switch the geometry, scale, readout, and IB factors separately; they are defined and evaluated in Section 3.4.

Hyperparameter selection. Each method selects its hyperparameters on the validation set from a shared grid: a six-value log-spaced β grid $\{10^{-4}, \ldots, 10\}$ for every variational baseline, and, for GPB and GPB-L, an additional three-value anchor-scale grid $\{1, 6, 12\}$ (6×3 combinations). The main table reports, per method and setting, the test metric of the best validation configuration, so all methods are compared under an identical selection protocol (Base has no sweep and early-stops as usual).

Scope of the theoretical comparison. The quantitative alignment bound in Theorem 2.1, $\mathbb{E}[D(Y)] \leq (C_{\mathrm{align}} + \varepsilon)/\beta$, applies only to a data/model pair satisfying the theorem's deterministic-label and realizability assumptions. We do not assume or verify those premises for the real-data benchmarks (including Cora, IMDb, and AG News), where label ambiguity, finite model capacity, and restricted prediction heads may be present. The benchmark experiments therefore target two empirical questions: whether the declared prior geometry transfers to the posterior, and whether the tied geometric readout remains predictive. They should not be read as direct tests of the theorem's $O(1/\beta)$ upper bound. A separate controlled experiment with strictly deterministic, explicitly realizable labels isolates the theorem's assumptions from benchmark realities; it is reported in Section 3.2.

β	fixed $a=6$: $\widehat{D}$	fixed $a=6$: $\beta\widehat{D}$	trainable $a=6$: $\widehat{D}$	trainable $a=6$: $\beta\widehat{D}$
10^{-3}	$4.9996\times10^{-3}\pm1.70\times10^{-3}$	5.2×10^{-6}	$7.7718\times10^{-3}\pm2.62\times10^{-3}$	7.8×10^{-6}
10^{-2}	$6.176\times10^{-4}\pm1.40\times10^{-4}$	6.0×10^{-6}	$8.356\times10^{-4}\pm3.49\times10^{-4}$	8.6×10^{-6}
10^{-1}	$1.9\times10^{-5}\pm8.0\times10^{-6}$	2.0×10^{-6}	$2.94\times10^{-5}\pm5.03\times10^{-6}$	3.0×10^{-6}
1	$< 10^{-6}$	$< 10^{-6}$	$0.6\times10^{-6}\pm0.5\times10^{-6}$	0.6×10^{-6}
10	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$

Table 1: Controlled exact-realizability experiment, selected β values. Means and standard deviations are over five seeds; entries below 10^{-6} reflect the six decimal places stored in the result CSV. The complete nine-point grid, including $a = 12$, is shown in the accompanying plots and is available in the experiment CSV.

3.2 Controlled realizability validation

To isolate the theorem's assumptions from data ambiguity and finite-task effects, we also ran the prescribed controlled experiment on a strictly deterministic synthetic classification task. We use $K = 10$ balanced classes with $x = e_y \in \mathbb{R}^{10}$, so $Y = f(X)$ and a linear encoder can represent the class-constant aligned posterior. The OPB posterior has $\tau^2 = 1$, with $a \in \{6, 12\}$, and we compare a fixed orthogonal frame with the trainable stateless polar frame. The CSV contains nine β values (10^{-3} through 10) and five seeds per configuration (180 runs total); all runs completed without a failure flag. We report the empirical $\widehat{D}$ defined in Eq. equation 15, together with $\beta\widehat{D}$, the same anchor-energy predictor's accuracy, and frame diagnostics.

For the aligned comparison distribution, Monte Carlo estimates of the aligned cross-entropy are $C_{\mathrm{G}} = 2.84 \times 10^{-4}$ for $a = 6$ (MC standard error 8×10^{-6}, using 10^7 samples) and zero at the reported precision for $a = 12$ (zero error events in the same sample budget; this is an empirical upper precision statement, not an assertion that the exact cross-entropy is zero). The task is so easily realizable that both deterministic and MC-10 test accuracy are 1.000 for every configuration. For $a = 6$, $\widehat{D}$ decreases from $4.9996 \times 10^{-3} \pm 1.70 \times 10^{-3}$ to below the CSV precision at the largest β for the fixed frame, and from $7.7718 \times 10^{-3} \pm 2.62 \times 10^{-3}$ to below precision for the trainable frame. The corresponding $\beta\widehat{D}$ remains at most 6.6×10^{-6} (fixed) and 9.4×10^{-6} (trainable), well below C_{G}. For $a = 12$, all reported $\widehat{D}$ and $\beta\widehat{D}$ values are zero to six decimal places for both frame choices. A linear fit of the nonzero $a = 6$ means against $1/\beta$ has a small absolute slope, 5.1×10^{-6} (fixed frame) or 7.9×10^{-6} (trainable frame), while the estimates reach a numerical floor. Thus the controlled run demonstrates rapid decrease of mismatch and satisfies the $\beta\widehat{D} \leq C_{\mathrm{G}}$ comparison at the reported precision, but it does not establish an exact $1/\beta$ law.

The geometric diagnostics are also consistent with the construction: anchor distances are $\sqrt{2}a$ (8.485 for $a = 6$ and 16.971 for $a = 12$), posterior minimum inter-class distances are within 0.1 of these values at the largest β, and the mean posterior center cosine is within 0.005 of zero. The fixed frame has minimum singular value 1; every trainable-frame run remains full rank, with the smallest recorded singular value 0.0585. These observations provide a controlled check of the conditional alignment mechanism. They do not establish the bound on real-world benchmarks, and because L^* and the theorem's ε are not known exactly, we do not claim a formal experimental verification of $\widehat{D} \leq (C_{\mathrm{G}} + \varepsilon)/\beta$.

3.3 Main results

Table 2 reports test accuracy (classification) and test R^2 (regression), averaged over runs, for each method's best configuration; bold marks the best mean per row and underline the second best, and the vertical rule separates GPB and its free-head ablation GPB-L from the baselines. NCBD (classification settings only) and AdaCap (regression settings only) are external non-IB baselines under the same protocol; dashes mark the settings where a method does not apply. Statistical claims (best vs. tied) follow the paired-difference confidence intervals of Appendix A.6. Standard deviations are omitted for width and are reported in Appendix A.5. Three observations.

The constructed geometry is free, and it usually pays. GPB is best or tied-best in 7 of 12 settings with mean rank 1.83, against 4.83 for the plain backbone and 5.50–7.00 for the variational baselines,

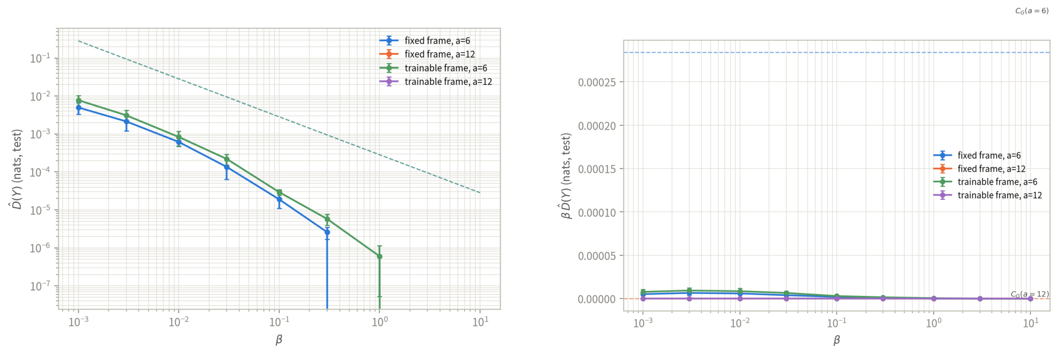

Figure 1: Controlled exact-realizability validation on the synthetic one-hot task. (a) Empirical anchor mismatch $\widehat{D}$ versus β; the dashed line is the C_{G}/β reference for $a = 6$. (b) $\beta\widehat{D}$ versus β with the corresponding C_{G} horizontal references. Curves show fixed and trainable frames at $a \in \{6, 12\}$; points are means and error bars are standard deviations over five seeds.

Table 2: Test accuracy (%) on classification tasks and test R^2 on regression tasks (mean over runs). Each entry reports the best configuration over the tuned β (and anchor-scale) grid; the best entry per dataset is bolded. NCBD (classification settings only) and AdaCap (regression settings only) are external non-IB baselines; dashes mark the settings where a method does not apply.

Dataset	Backbone	Base	VIB	SVIB	NIB	CEB	DVCCA	NCBD	AdaCap	GPB	GPB-L
MNIST	MLP	98.33	98.28	98.15	98.27	98.43	98.24	97.95	–	**98.54**	98.43
MNIST	CNN	99.13	99.28	99.23	99.22	99.30	99.29	99.18	–	**99.32**	99.31
ImageNet-100	MLP	83.12	82.84	83.02	82.35	82.77	82.99	83.64	–	**84.40**	84.01
ImageNet-100	CNN	83.40	82.94	83.11	82.91	82.97	83.21	80.56	–	**86.08**	84.88
Cora	GCN	87.68	86.83	86.46	86.35	86.27	86.90	87.34	–	**88.12**	87.86
IMDb	RNN	84.85	85.50	85.99	86.14	85.55	85.69	86.69	–	86.84	**86.99**
AG News	RNN	**92.94**	92.72	92.86	92.91	92.83	92.78	92.37	–	92.87	92.92
Cal. Housing	MLP	0.7618	0.7901	0.7897	0.7862	0.7894	0.7878	–	0.7758	0.7895	**0.7914**
STS-B	RNN	**0.2899**	0.2719	0.2697	0.2417	0.2668	0.2713	–	0.1613	0.2762	0.2755
ZINC	GCN	0.6348	0.6339	0.6325	0.6178	0.6257	0.6294	–	**0.6770**	0.6344	0.6323
AgeDB	MLP	0.2180	0.3173	0.3385	0.2710	0.3317	0.3180	–	0.3482	**0.3536**	0.3320
AgeDB	CNN	0.3914	0.3342	0.3323	0.3337	0.3882	0.3205	–	0.4684	**0.4700**	0.4388
Mean rank		4.83	5.75	5.50	7.00	5.96	6.00	6.43	4.40	1.83	2.54
Best or tied-best		2	0	0	0	0	0	0	1	7	2

none of which is ever best; among the external baselines NCBD never wins and AdaCap wins once (ZINC). By the paired bootstrap 95% CIs (Appendix A.6), the GPB–CEB difference is significantly positive in 8 of 12 settings and within the non-inferiority margin in three more; only on STS-B does the interval overlap zero beyond the margin ($[-0.009, +0.025]$), so no setting shows a significant CEB advantage. GPB dominates CEB by mean in all 12 settings and is best among all methods on seven rows. The gains concentrate where the label structure is rich or supervision is scarce: $+3.1$ accuracy over CEB on ImageNet-100 with the CNN backbone (86.1 vs. 83.0), $+1.9$ on Cora, and $+0.08$ R^2 on AgeDB over the plain CNN (0.470 vs. 0.388). On saturated settings (MNIST) the margin is small but consistent, and on the two rows where the plain backbone remains best (STS-B, AG News) the gap is marginal (≤ 0.03): the geometry is neutral there rather than harmful. On ZINC every bottleneck method trails the plain backbone, but the closed-form AdaCap readout edges it out (0.677 vs. 0.635 R^2), the one setting where any comparison method significantly beats GPB (see below).

Prediction does not require the free head. GPB-L, the free linear head on the same geometric prior, is best or tied-best in 2 of 12 settings (mean rank 2.54), and the GPB–GPB-L gap is within 0.4 accuracy or R^2 points in 11 of 12 settings (the exception is ImageNet-100 CNN, 1.2 points): the zero-parameter anchor-energy classifier — whose weights are the frame itself — reaches the accuracy of a freely trained head in almost all settings, so the degrees of freedom removed by construction are not required for prediction, as claimed in the introduction. The three settings where

GPB-L edges out GPB by mean (IMDb, AG News, Cal. Housing) do so with paired CIs overlapping zero, confirming that the tie does not inflate the numbers. On the five regression rows the tied projection head of EPB wins or matches the free regression head in four (the exception is Cal. Housing by 0.002).

The tie regularizes the high-cardinality and low-data regimes. The two settings with the largest GPB margins (ImageNet-100, with $K = 100$, and AgeDB, a small, high-variance regression task) are exactly those where an unconstrained classifier can overfit its own scale or the scarce supervision: fixing the prediction rule to the prior's geometry removes both escape routes, consistent with the fixed-scale condition under which the aligned surrogate of Proposition A.4 has a unique interior optimum per β.

External baselines do not close the gap. By the paired CIs (Appendix A.6), GPB beats NCBD significantly on 6 of 7 classification settings; the exception is IMDb (+0.2 points, CI $[-0.5, +0.7]$ crossing zero), where the GPB margin is within noise. The largest margin is ImageNet-100 with the CNN backbone (+5.5 points, CI $[+4.3, +6.9]$). On the regression side GPB beats AdaCap significantly on Cal. Housing ($+0.014$ R^2) and STS-B ($+0.115$ R^2), and the AgeDB intervals straddle zero (the CNN one within the non-inferiority margin). The single reversal is ZINC, where AdaCap is the only method with a significant advantage over GPB (-0.043 R^2, CI $[-0.046, -0.041]$): its ridge readout, refit on the training set at evaluation time, exploits the low-dimensional regression structure of molecular-graph labels without any compression objective. Neither external baseline performs information-bottleneck compression, so their sweeps exercise the same data but a different axis of the comparison.

3.4 Attribution ablations: what drives the gain?

The GPB construction bundles three mechanisms — the orthogonal/isometric geometry, the fixed anchor scale, and the tied geometric readout — with the IB objective itself, and the main table alone cannot attribute the gain among them. This section switches each factor off separately. All variants are trained under the protocol of Section 3.1 (same β and anchor grids, five seeds, per-variant configuration selection) on the three settings of the mechanism analysis, and the reference models (Base, CEB, GPB, GPB-L) are re-trained in the same batch so every comparison is paired by seed.

The ablation ladder. *NCM-learn* removes the bottleneck entirely: $z = \mu_\phi(x)$ is deterministic, prediction is the energy rule $\mathrm{logit}_k = -\|z - m_k\|^2/2$ on trainable prototypes m_k, and there is no posterior and no KL. *NCM-ortho* fixes the prototypes to the orthogonal frame $a\, e_k$: fixed geometry, still no IB. *CEB-energy* and *CEB-tied* mount the same energy readout (resp. the tied projection) on CEB's trainable free reference: free geometry, free scale, restricted readout, with IB. *OPB-FS* and *EPB-FS* keep the orthogonal frame and the energy (resp. tied) readout but make the anchor radius learnable — per-class scales s_k, or a scalar ρ — initialized at the anchor scale, so only the fixed scale is switched off. Together with Base, CEB, GPB-L and GPB the ladder switches each factor independently: geometry (CEB-energy vs. OPB-FS), scale (OPB-FS vs. GPB), readout (GPB-L vs. GPB; CEB vs. CEB-energy), and the IB term itself (NCM-ortho vs. GPB).

Where the gain comes from. Table 3 reports the best configuration of each variant and Table 4 the factor-wise paired differences. On ImageNet-100 the decomposition is clean: the geometry contributes $+2.67$ points (OPB-FS $-$ CEB-energy, CI $[2.45, 2.92]$), the IB term $+1.89$ (GPB $-$ NCM-ortho, $[1.41, 2.36]$), and the restricted readout $+0.82$ (GPB $-$ GPB-L, $[0.34, 1.17]$), while the fixed scale is neutral (-0.04, $[-0.07, -0.01]$). The readout effect flips sign on the free-geometry side, where the energy head costs CEB 1.19 points ($[-1.50, -0.96]$): the restricted readout pays only on the constructed geometry. Most informative is the negative result on the prototype readout alone: NCM-learn trails the plain backbone by 0.96 points ($[-1.30, -0.69]$), so the gain is not a nearest-class-mean effect in disguise — the prototypes need the IB objective and the orthogonal geometry to carry prediction. On MNIST the same ordering holds with small margins (geometry $+0.32$, IB $+0.34$, scale $+0.09$, readout ≈ 0), and on Cal. Housing every EPB variant ties within 0.01 R^2, consistent with the flat EPB plateaus of Section 3.5. The conclusion is that the orthogonal geometry is the primary driver of the ImageNet-100 gain, and it acts *within* the IB framework: geometry, IB,

variant	MNIST (Acc %)	ImageNet-100 (Acc %)	Housing (R^2)
Base	$98.3_{\pm 0.1}$	$83.1_{\pm 0.1}$	$0.762_{\pm 0.024}$
NCM-learn	$98.2_{\pm 0.2}$	$82.2_{\pm 0.4}$	–
NCM-ortho	$98.1_{\pm 0.1}$ ($a = 1$)	$82.3_{\pm 0.6}$ ($a = 1$)	–
CEB	$98.4_{\pm 0.1}$ ($\beta = 0.01$)	$82.8_{\pm 0.2}$ ($\beta = 0.01$)	$0.789_{\pm 0.002}$ ($\beta = 0.0001$)
CEB-energy	$98.2_{\pm 0.1}$ ($\beta = 0.001$)	$81.6_{\pm 0.2}$ ($\beta = 0.001$)	–
GPB-L	$98.4_{\pm 0.1}$ ($\beta = 0.1$, $a = 12$)	$83.4_{\pm 0.7}$ ($\beta = 1$, $a = 12$)	–
GPB	$98.4_{\pm 0.1}$ ($\beta = 0.1$, $a = 12$)	$84.2_{\pm 0.1}$ ($\beta = 1$, $a = 12$)	–
OPB-FS	$98.5_{\pm 0.0}$ ($\beta = 0.1$, $a = 12$)	$84.3_{\pm 0.2}$ ($\beta = 1$, $a = 12$)	–
CEB-tied	–	–	$0.781_{\pm 0.006}$ ($\beta = 0.01$)
EPB-L	–	–	$0.791_{\pm 0.003}$ ($\beta = 0.0001$, $a = 6$)
EPB	–	–	$0.788_{\pm 0.004}$ ($\beta = 0.0001$, $a = 6$)
EPB-FS	–	–	$0.787_{\pm 0.008}$ ($\beta = 0.001$, $a = 6$)

Table 3: Attribution ablation: best-configuration test metrics (mean $\pm$ std over five seeds). Configurations are selected by the mean test metric, applied identically to every variant, so all comparisons remain paired by seed; the selection is slightly optimistic in level but shared across the table. One OPB-FS configuration (ImageNet-100, $\beta = 10^{-4}$, $a = 1$) diverged during training (free scales grow without penalty at negligible β) and is excluded.

factor	comparison	MNIST	ImageNet-100	Housing
readout	GPB − GPB-L	$+0.02$ $[_{-0.05,\ 0.08}]$	$+0.82$ $[_{0.34,\ 1.17}]$	–
readout (EPB)	EPB − EPB-L	–	–	-0.00 $[_{-0.01,\ -0.00}]$
readout (free geometry)	CEB-energy − CEB	-0.21 $[_{-0.30,\ -0.12}]$	-1.19 $[_{-1.50,\ -0.96}]$	–
geometry	OPB-FS − CEB-energy	$+0.32$ $[_{0.26,\ 0.40}]$	$+2.67$ $[_{2.45,\ 2.92}]$	–
geometry (EPB)	EPB-FS − CEB-tied	–	–	$+0.01$ $[_{-0.00,\ 0.01}]$
fixed scale	GPB − OPB-FS	-0.09 $[_{-0.16,\ -0.02}]$	-0.04 $[_{-0.07,\ -0.01}]$	–
fixed scale (EPB)	EPB − EPB-FS	–	–	$+0.00$ $[_{-0.01,\ 0.01}]$
IB increment*	GPB − NCM-ortho	$+0.34$ $[_{0.22,\ 0.45}]$	$+1.89$ $[_{1.41,\ 2.36}]$	–
prototype structure	NCM-ortho − NCM-learn	-0.06 $[_{-0.20,\ 0.11}]$	$+0.16$ $[_{-0.49,\ 0.82}]$	–
prototype readout alone	NCM-learn − Base	-0.16 $[_{-0.28,\ -0.05}]$	-0.96 $[_{-1.30,\ -0.69}]$	–

Table 4: Factor-wise paired differences (accuracy points on MNIST and ImageNet-100, R^2 on Housing; 95% bootstrap percentile CIs, $B = 10{,}000$, fixed seed, paired across the shared five seeds at each variant's best configuration; positive values favor the first term). *NCM-ortho fixes the frame at $a\, e_k$ while GPB trains it, so the IB comparison additionally contains frame trainability.

and readout each contribute positively, in that order, and none of the bundled mechanisms is doing the work alone.

3.5 Compression–accuracy and the KL certificate over β

The main table selects the best (β, a) per method; the figures below report the compression–accuracy trade-off on one classification and one regression setting (ImageNet-100 and Cal. Housing, both with the MLP backbone). Each setting has two companion panels over the same β grid: the task metric against β (panels a–b), and the realized test-set $\mathbb{E}[\mathrm{KL}]$ — the certificate itself — against β (panels c–d). The certificate panels are what make a β-matched sweep interpretable: the two methods reach different compressions at the same β — on Housing at $\beta = 0.1$, CEB sits at $\mathbb{E}[\mathrm{KL}] \approx 0.09$ while EPB ($\rho = 12$) is at ≈ 0.27 — so the compression each point actually achieves is read off directly rather than assumed from β. No points are omitted: collapsed references and variance explosions are visible at the extremes of the sweep.

Figure 2 exhibits the two roles of the anchor scale across the β sweep, with panels (c)–(d) translating each point into its realized compression. *(i) Without a scale, the reference collapses.* On ImageNet-100, CEB's accuracy erodes to 71.5% at $\beta = 0.1$ ($\mathbb{E}[\mathrm{KL}] \approx 3.6$) and its reference escapes entirely at $\beta \geq 0.5$ ($\mathbb{E}[\mathrm{KL}] \to 0$ with accuracy at chance, panel c); OPB at $a = 1$ survives one grid step longer (83.3% at $\beta = 0.1$) and collapses at the same end of the sweep. OPB ($a = 6$) holds accuracy above 74% out to $\beta = 5$ and falls to 64.7% at $\beta = 25$. On Housing, CEB drops to $R^2 = 0.49$ at $\beta = 0.1$ and below 0.1 by $\beta \geq 0.5$, while every EPB curve keeps $R^2 \geq 0.76$ out to $\beta = 25$. Orthogonality alone therefore does not prevent the collapse: the frame is fixed up to rotation, but the

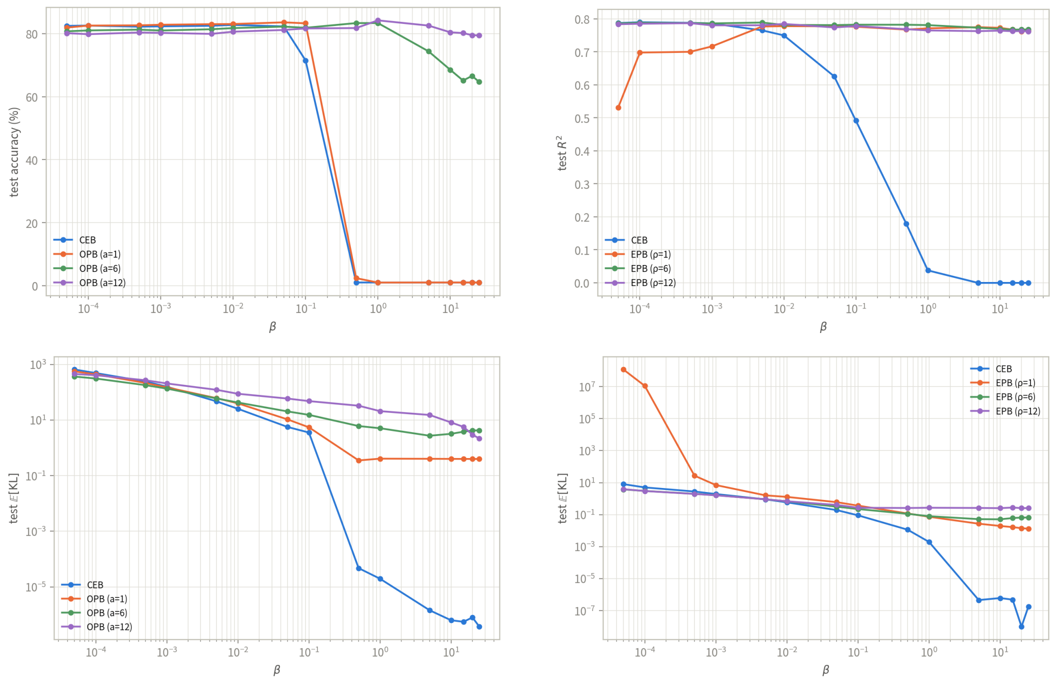

Figure 2: Compression–accuracy and the realized certificate over β (log scale), $\beta \in \{5{\cdot}10^{-5}, \dots, 25\}$; CEB one curve per panel, GPB three (a on classification, ρ on regression, both $\in \{1, 6, 12\}$). (a)–(b) Test accuracy (%) on ImageNet-100 (MLP) and test R^2 on Cal. Housing (MLP) against β: OPB with $a = 12$ holds the plateau out to $\beta = 25$ while CEB and OPB ($a = 1$) collapse. (c)–(d) The realized test-set $\mathbb{E}[\mathrm{KL}]$ against β (log scale) for the same runs: CEB's certificate collapses with its reference, EPB ($\rho = 1$) explodes near $\beta = 5{\cdot}10^{-5}$, and the larger anchors keep $\mathbb{E}[\mathrm{KL}]$ in a stable band.

anchor-energy classifier's margin is a^2/τ^2 (Proposition 2.2), and at $a = 1$ that margin is too small to carry a solution at strong compression. *(ii) The anchor scale controls where the collapse sets in, and at matched compression the geometry pays or ties.* On ImageNet-100, OPB ($a = 12$) attains 84.2% at $\beta = 1$ ($\mathbb{E}[\mathrm{KL}] \approx 21$), above CEB's best point anywhere on its sweep (82.8% at $\beta = 10^{-2}$), and still 79.4% at $\beta = 25$ ($\mathbb{E}[\mathrm{KL}] \approx 2.2$); on Housing the four curves peak within half a point (CEB 0.789, EPB 0.786–0.789), but only EPB retains its peak into the strong-compression regime: the plateau is the flat EPB segment where R^2 stays above 0.76 from $\beta = 0.1$ out to $\beta = 25$ while the certificate keeps shrinking.

A visible cost remains in the weak-compression regime: on ImageNet-100 at $\beta = 10^{-4}$ ($\mathbb{E}[\mathrm{KL}] \gtrsim 300$), OPB trails CEB by one to three points ($a = 6$: 81.0 vs. 82.6; $a = 12$: 79.8 vs. 82.6): the price of the fixed-scale energy rule when compression is weak. On Housing the same regime shows EPB ($\rho = 1$) dipping to $R^2 = 0.53$ at $\beta = 5{\cdot}10^{-5}$, where its certificate explodes to $\approx 10^8$ (panel d): with a unit axis and negligible β the off-axis variance is unconstrained. The construction pays for itself in the strong-compression regime, where CEB is at chance.

The mechanism is consistent with the conditional analysis. At strong β the posterior is empirically close to the aligned solution. In the idealized realizable setting, Theorem 2.1(a) bounds the expected anchor mismatch by $(\ln K + \varepsilon)/\beta$, and the accuracy achievable at alignment is the aligned Gaussian mixture's accuracy, governed by the scale-to-noise ratio a/τ (the nearest-anchor error probability is $\Phi(-a/(\sqrt{2}\tau))$; appendix). For the benchmark runs we do not assume the theorem's realizability premises; the realized KL and geometry are inspected directly. With the classifier scale fixed at a/τ^2 by construction, the only remaining lever at full compression is a itself: the anchor scale converts the compression knob from a collapse dial into a plateau dial. The KL decomposition shows the two channels at work: OPB's $\mathbb{E}[\mathrm{KL}]$ is dominated by the mean-mismatch term (anchor displacement; $a = 12$, $\beta = 1$: 20.6 mean vs. 0.36 variance), while CEB's reference absorbs compression by

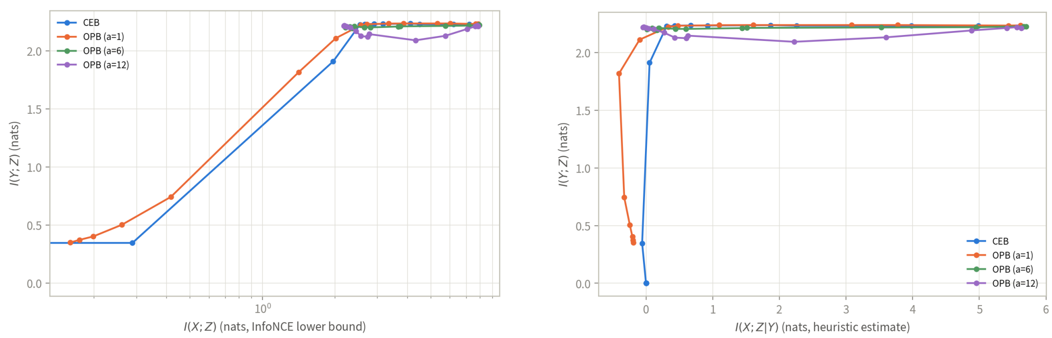

Figure 3: Information plane (a) and certificate plane (b) on MNIST over the β sweep, five runs per point, lines connect β in ascending order; x is log-scaled in (a), and the certificate axis in (b) switches to a linear scale wherever the estimated residual turns negative (OPB $a = 1$ at $\beta \geq 5$, see text). CEB one curve per panel; OPB three ($a \in \{1, 6, 12\}$). CEB follows the classic corner — compression and task information drop together past a critical $\beta = 0.5$ with bimodal runs — and collapses to chance by $\beta = 1$; OPB ($a = 6$, 12) keeps accuracy flat out to $\beta = 25$ with $I(X;Z)$ saturating at ≈ 2.2–2.4 nat, essentially all of it task-relevant (certificate ≈ 0). The Cal. Housing panels are deferred to Fig. 8 in the appendix.

shrinking both terms toward zero: the reference chases the posterior, exactly the degeneracy the constructed geometry excludes (Lemma A.1). This is the empirical counterpart of the mismatch-dependent separation bounds (Proposition 2.4; corollaries in the appendix) and of the fixed-scale condition of Proposition A.4; consistent with that proposition's scope, we do not claim that the map $\beta \mapsto$ accuracy is monotone, only that the constructed scale keeps the plateau from collapsing.

3.6 Information-plane analysis

The certificate tells us *which* gap the KL measures; the panels below inspect *what* each method actually discards. On every sweep checkpoint we estimate the three quantities of the canonical information plane (Fig. 3): $I(X;Z)$ by the InfoNCE lower bound (van den Oord et al., 2018) ($\log B - \mathcal{L}_{\mathrm{NCE}}$, $B = 4096$, one critic per run trained on the training split, the bound read off the test split with $z \sim q(z|x)$); $I(Y;Z)$ by the classification lower bound $H(Y) - \mathrm{CE}(\text{logits})$; and the residual $I(X;Z|Y)$ as the difference of the two — a heuristic, not a bound. The true residual is nonnegative by the data-processing inequality; negative plotted values mean the estimators, not the theory, gave way, and they occur exactly where the InfoNCE bound fails (see the caveats below). Consistent with Remark A.4, this section reports estimator-level observations on the sweep checkpoints and claims nothing about the exact plane.

(i) The corner is sharp. On MNIST, CEB moves from $I(X;Z) = 2.53$ at 98.2% ($\beta = 0.1$) to 1.96 at 86.0% ($\beta = 0.5$) to 0.29 at 26.8% ($\beta = 1$), and to chance beyond. The $\beta = 0.5$ point is a critical point rather than a midpoint: the per-seed accuracy is bimodal ($86.0 \pm 22.5\%$; one run collapses to 41% while four sit at 96.5–97.8%). The two axes fall together — the reference absorbs the compression by absorbing the task information.

(ii) The plateau has a floor at the task information. OPB ($a = 6$, 12) keeps MNIST accuracy at 98.1–98.6% for every β up to 25; $I(X;Z)$ saturates at 2.2–2.4 nat while $I(Y;Z)$ stays at ≈ 2.2 (against $H(Y) = \ln 10 \approx 2.30$), and the certificate sits at ≈ 0 ($a = 12$, $\beta = 15$: 0.00 ± 0.21; $a = 6$, $\beta = 5$: 0.09). The floor is the data-processing inequality made visible: at full compression the frame leaves room for exactly the task-relevant information and nothing else.

(iii) At matched compression the geometry keeps more of the task. Table 5 pairs each CEB point with the OPB configuration of nearest estimated $I(X;Z)$. At ≈ 2.0 nat, CEB ($\beta = 0.5$) retains 86.0% while OPB ($a = 1$, $\beta = 0.5$) retains 97.8%; at ≈ 2.5 nat ($\beta = 0.1$) the two tie (98.2% vs. 98.1%) — the geometry pays nothing where compression is mild and carries the task where it is not.

Estimator caveat. One region must be read with care, and it is not used for any claim above: OPB ($a = 1$) at $\beta \geq 5$. There the plotted estimates collapse ($I(X;Z) \leq 0.42$ nat) while accuracy stays

	CEB		OPB (nearest $I(X;Z)$)		
Setting	$I(X;Z)$	Acc	config	$I(X;Z)$	Acc
MNIST, $\beta = 0.1$	2.53 ± 0.03	98.2 ± 0.1	$a = 1, \beta = 0.1$	2.55 ± 0.01	98.1 ± 0.1
MNIST, $\beta = 0.5$	1.96 ± 0.53	86.0 ± 22.5	$a = 1, \beta = 0.5$	2.01 ± 0.03	97.8 ± 0.1

Table 5: Matched-compression comparison on MNIST: each CEB checkpoint is paired with the OPB configuration of nearest estimated $I(X;Z)$ (nats; mean $\pm$ std over five runs). At matched compression the geometric prior retains strictly more of the task; the CEB $\beta = 0.5$ entry is the bimodal critical point of the collapse transition (individual runs at 97% or 41%).

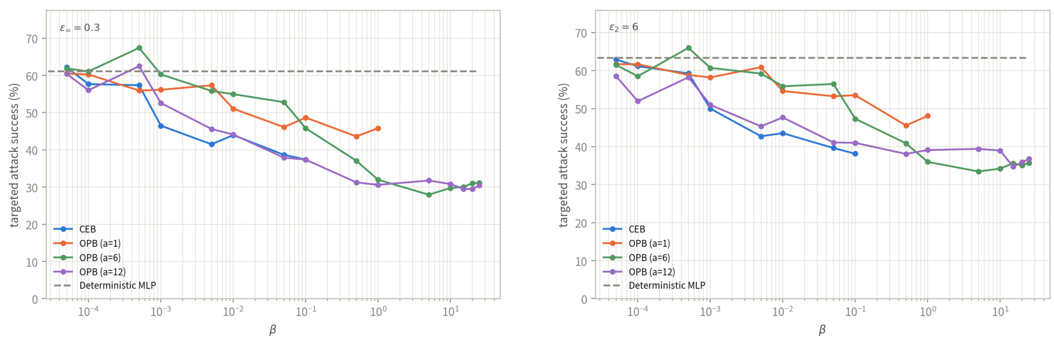

Figure 4: Targeted PGD attack success rate against β (log scale) on MNIST (MLP backbone): target class 0, 20 steps, EOT-10 gradients and 10-sample MC predictions on the stochastic path. (a) L_∞, $\varepsilon = 0.3$; (b) L_2, $\varepsilon = 6.0$. The dashed grey line is the deterministic MLP baseline. Every curve is drawn only over β values with clean accuracy above the shared 90% failure threshold (CEB: $\beta \leq 0.1$; OPB $a = 1$: $\beta \leq 1$; $a = 6$, 12: full grid). Every curve falls with β, and OPB keeps gaining robustness across its valid range: at $\beta = 25$, OPB ($a = 6$, 12) reach $31.1\%/30.5\%$ — about 30 points below the baseline.

97.6–97.8%; by Fano's inequality the 2.3% error alone requires $I(Y;Z) \geq H(Y) - h(0.023) - 0.023 \ln 9 \approx 2.1$ nat, so the leftward spur is a bound failure, not compression: z quantizes onto the ten anchors, near-duplicate positives saturate the InfoNCE objective (Poole et al., 2019), and overconfident errors inflate the cross-entropy.

3.7 Adversarial robustness

We evaluate robustness to white-box attacks following the CEB paper's *targeted* variant (its Figure 4, bottom panels): targeted projected gradient descent (PGD, 20 steps, step size $2.5\varepsilon/\text{steps}$, random start) toward a single class, with success measured as the fraction of originally correct predictions flipped to that class. The target is digit 0, the most distinctive MNIST class (the analogue of Fashion-MNIST "trouser"), so that class confusion does not masquerade as robustness. ε is defined in the normalized pixel space (MNIST mean/std); we attack 1,000 test images and report the mean and standard deviation over runs. All stochastic models are attacked and evaluated on their *stochastic* path: each PGD step averages gradients over 10 samples of $z \sim q_\phi(z|x)$ (expectation over transformation), and every prediction is the majority class under 10 MC samples. We use $\varepsilon_\infty = 0.3$ and $\varepsilon_2 = 6.0$ — the smallest L_2 budget we tested at which the targeted attack engages the decision boundaries at all (at 2.0 it cannot reach the target region from any other class and sits at a near-zero floor for every model). Every curve is drawn only over the β values at which the model's clean accuracy clears a shared 90% failure threshold under the same stochastic protocol: below it a configuration is counted as failed, and a failed model's near-zero attack success is not robustness. The threshold binds CEB at $\beta \leq 0.1$ (beyond, the reference collapses to chance) and OPB ($a = 1$) at $\beta \leq 1$ (beyond, its posterior saturates at the fixed prior variance while its means pin to the radius-1 anchors, and the MC-sampled predictions fall to 0.59–0.84 clean accuracy even though the deterministic path keeps 0.98); OPB ($a = 6$, 12) clears the threshold across the full grid.

Figure 4 shows the targeted-attack picture. Every curve falls as β grows, reproducing the CEB paper's compression–robustness gradient under a protocol the deterministic path cannot game. Within

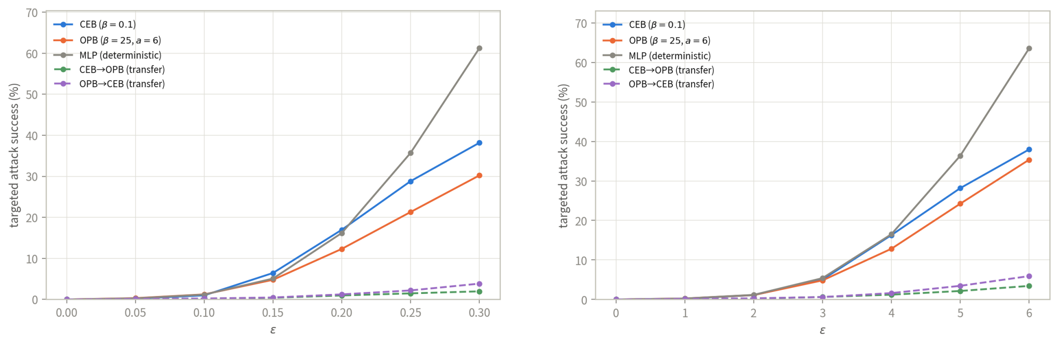

Figure 5: Transfer attacks (gradient-masking check) on MNIST (MLP backbone): targeted PGD adversaries generated on one model and evaluated on another, against ε. (a) L_∞; (b) L_2. Solid: white-box self-attacks of CEB ($\beta = 0.1$), OPB ($\beta = 25$, $a = 6$) and the deterministic MLP baseline. Dashed: transfer attacks. CEB→OPB stays an order of magnitude below OPB's own white-box attack, so OPB's extreme-compression robustness is not gradient masking; OPB→CEB transfers little as well — adversarial directions found on the strongly compressed posterior are largely model-specific.

CEB's operating range the two geometries are comparable at matched β (at $\beta = 0.1$ under L_∞: 37.4% for both CEB and OPB, $a = 12$), but the β axis again hides the compression gap — at $\beta = 0.1$ CEB is the *more* compressed of the two (realized $\mathbb{E}[\mathrm{KL}] = 0.68$ vs. 1.23 for OPB, $a = 12$). At compression-matched points the two geometries are at par within run-to-run variation: OPB ($a = 12$, $\beta = 1$; $\mathbb{E}[\mathrm{KL}] = 0.55$) attains 30.6% vs. CEB's 37.4% under L_∞, a gap inside the seed spread. The decisive margin is the operating range. Under the shared 90% failure threshold, CEB's curve ends at $\beta = 0.1$ and OPB ($a = 1$)'s at $\beta = 1$, while the larger anchors keep accuracy at ≈ 0.98 out to $\beta = 25$, where their targeted success has fallen to 31.1%/30.5% ($a = 6/12$) under L_∞ and 35.7%/36.8% under L_2 — about 30 points below the deterministic baseline and below anything CEB attains with its accuracy intact. The $a = 1$ truncation is itself informative: at $\beta \geq 5$ the small anchor's posterior saturates at the fixed prior variance (the KL variance-mismatch minimum) while its means pin to the radius-1 anchors, so the stochastic path the robustness protocol evaluates on drowns in noise — the failure mode the larger anchors avoid. We do not over-claim: this is one dataset and one target class.

Transfer attacks (gradient-masking check). A white-box attack could fail simply because the target model's gradients are degenerate, crediting dead models with robustness. We follow the CEB paper's transfer protocol as the check: generate targeted PGD adversaries on one model (the same 20-step, EOT-10 recipe) and evaluate them on another (10-sample MC predictions), sweeping ε over 0.05–0.3 (L_∞) and 1–6 (L_2). The models are CEB ($\beta = 0.1$, the last point of its operating range), OPB at maximum compression ($\beta = 25$, $a = 6$; realized $\mathbb{E}[\mathrm{KL}] = 0.40$), and the deterministic MLP baseline, whose white-box curve reaching 61% at $\varepsilon_\infty = 0.3$ confirms that the attack itself is not broken.

Figure 5 reads each transfer curve against its target's own white-box attack, not against the source. CEB→OPB tops out at 2.0% (L_∞) and 3.4% (L_2) — an order of magnitude below OPB's own white-box attack (30.2%/35.4%) — so the extreme-compression robustness of OPB survives adversaries that never touch its gradients: it is geometry, not gradient masking. The reverse direction transfers little as well ($\leq 3.8\%$/5.9%, against CEB's own white-box 38.2%/37.9%): adversarial directions found on the strongly compressed posterior are largely model-specific, so neither model's robustness is an artifact of the other's gradients.

3.8 Mechanism validation: the geometry drives representation and prediction

The main sweep established the predictive adequacy of the constructed geometry; here we verify the mechanism behind it, on the two representative settings of the analysis, MNIST (MLP, classification)

Table 6: Mechanism validation: the constructed geometry pulls the posterior into the orthogonal frame (classification) and onto the isometric axis (regression). Geometric concentration: mean off-diagonal cosine of the posterior class-center Gram matrix (classification) / off-axis residual fraction (regression). Scale following: class-center radius $\|\bar{c}\|$ against the anchor scale a (classification) / axial slope against ρ (regression). Cross-seed stability: std. over runs. Means $\pm$ std. over five runs.

Model (task)	Task	Geometric concentration	Scale following	Cross-seed stability
CEB (MNIST)	Classif.	0.1205$\pm$0.0026	5.14	0.0201
OPB (MNIST)	Classif.	0.0203$\pm$0.0021	11.00 ($a = 12$)	0.0082
CEB (Housing)	Regress.	0.0233	2.65$\pm$0.59	0.593
EPB (Housing)	Regress.	0.0002	9.22$\pm$0.15 ($\rho = 12$)	0.155

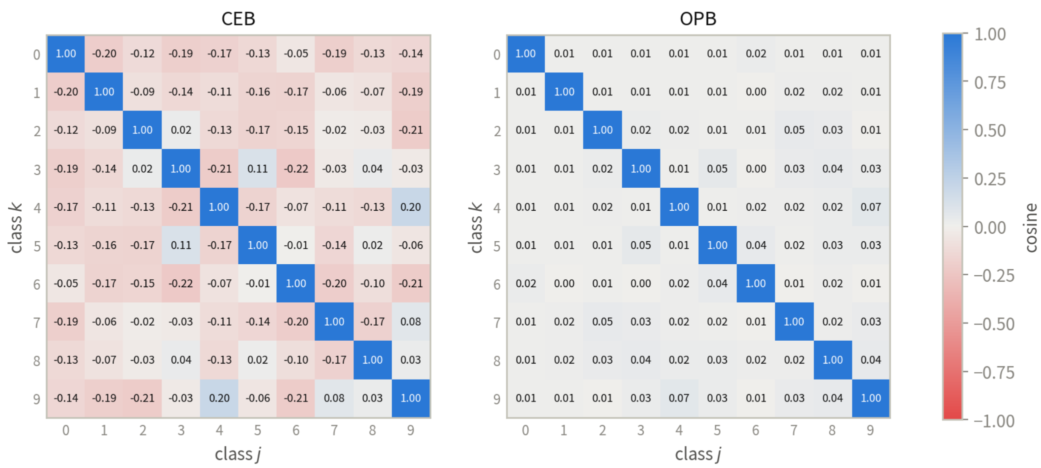

Figure 6: Pairwise-cosine Gram matrix of the posterior class centers on MNIST, averaged over five runs. OPB is nearly diagonal: the posterior centers are orthogonal; CEB shows no such structure.

and California Housing (MLP, regression), trained at the matched compression $\beta = 0.1$ with anchor scale $a = \rho = 12$.

The prior. CEB's reference is arbitrary and drifts across seeds: the mean off-diagonal cosine of its learned class-prior table is 0.119 ± 0.002, and the scale of its learned regression axis varies by 2.07 ± 0.88 across runs. The constructed prior has no such freedom: the OPB class priors are exactly orthogonal (off-diagonal cosine 0, pairwise distance $\sqrt{2}\,a = 16.97$), and the EPB axis is fixed at $\rho = 12.000$ with exact isometry through the origin, both verified at machine precision.

The posterior. Table 6 shows that this geometry transfers to the posterior. On classification the posterior class centers are nearly orthogonal (the mean off-diagonal Gram cosine is $5.9\times$ lower than CEB's) and sit at radius $\|\bar{c}\| = 11.00$ (92% of a), with $2.5\times$ lower cross-seed drift; the Fisher ratio is comparable (15.3 vs. 15.7, reported as is). On regression the posterior lives essentially *on* the isometric axis: the off-axis residual fraction is $141\times$ lower than CEB's, and the axial slope tracks ρ with $3.8\times$ lower cross-seed variation. The slope is 9.22 rather than $\rho = 12$ because the encoder's uncertainty shrinks the posterior along the axis (the slope equals $\rho \cdot R^2_{\text{axis}}$, as $12 \times 0.78 \approx 9.4$), and the axis *direction* itself can rotate rigidly with the representation (mean direction cosine 0.03–0.06 for both models), so the discriminative stability evidence is the scale, not the direction. Figures 6 and 7 show both mechanisms directly.

Prediction reads off the geometry. Both heads of Section 2.2 have no free parameters. At the matched configuration the anchor-energy classifier attains 98.44% test accuracy against CEB's 98.16% on MNIST, and the tied projection head attains $R^2 = 0.778$ against CEB's 0.491 on Housing: under this strong compression CEB collapses while the geometric readout retains the signal, consistent with the main-table GPB column (Section 3.3) and with the nearest-anchor analysis of the appendix. The mechanism chain is therefore complete: the constructed prior pulls the posterior

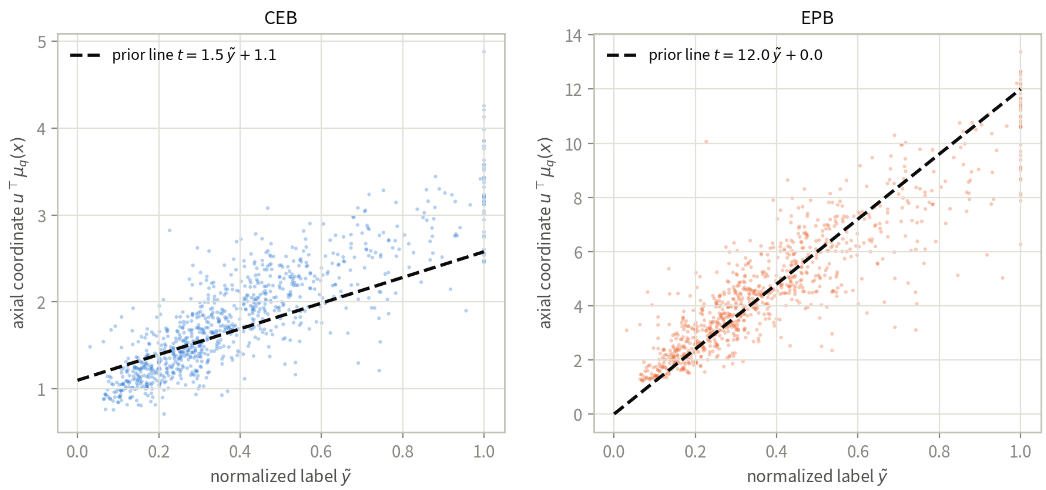

Figure 7: The axial coordinate $u^\top \mu_q(x)$ against the normalized label on Housing (one representative run per model). The EPB posterior hugs the prior line $\rho\tilde{y}$, shrunk toward $\rho\,\mathbb{E}[\tilde{y} \mid x]$ by the encoder's uncertainty.

into the declared geometry, and a zero-parameter rule that reads exactly that geometry carries the prediction.

4 Related Work

Information bottleneck and conditional compression. The information bottleneck (IB) principle formulates representation learning as retaining information about a target while discarding information about the input (Tishby et al., 2000). Deep VIB turns this principle into a neural objective by using a reparameterized stochastic encoder and a variational marginal prior (Alemi et al., 2017). Nonlinear and non-parametric bottleneck variants study alternative variational bounds and distance-based estimates of the information in a representation (Kolchinsky et al., 2019). The deterministic-scenario analysis of Kolchinsky & Tracey (2019) further shows that the usual IB multiplier can fail to sweep an interior information–accuracy trade-off when labels are deterministic; the squared-compression baseline in our experiments follows this line of work.

The conditional entropy bottleneck (CEB) replaces the marginal reference of VIB by a label-conditional reference and thereby targets conditional residual information (Fischer, 2020). Comparisons of variational bounds clarify when this conditional reference is tighter than a marginal one (Geiger & Fischer, 2020). CEB has also been applied to large-scale visual representations and robustness evaluations (Fischer & Alemi, 2020; Lee et al., 2021), while multivariate extensions retain the same variational bottleneck viewpoint (Abdelaleem et al., 2025). These methods leave the relationships among distinct conditional reference means unspecified. GPB keeps the CEB posterior and its single conditional KL, but restricts the reference family itself: OPB declares a class frame and EPB declares a label-distance-preserving axis.

Information bottleneck for robust representations. Several works use a bottleneck to suppress nuisance or non-robust information. The HSIC bottleneck adds layer-wise dependence penalties and analyzes adversarial robustness (Wang et al., 2021). Adversarial Information Bottleneck explicitly incorporates adversarial objectives (Zhai & Zhang, 2022), and an NLP-specific variant separates task-relevant robust features from manipulable features (Zhang et al., 2022). Causal Information Bottleneck uses a causal decomposition for the same robust-feature goal (Hua et al., 2022). These approaches modify the information criterion, feature selection, or training examples. In contrast, the geometric constraint in GPB is placed on the conditional prior and is evaluated with the ordinary CEB KL; the robustness experiments therefore test whether a declared prior geometry is transferred to the posterior rather than introducing adversarial training as an additional objective.

Geometric classification representations. Neural-collapse studies document an approximately simplex-ETF arrangement of class means and classifier directions near the terminal phase of training (Papyan et al., 2020; Zhu et al., 2021). The connection between neural collapse, supervised contrastive learning, and an information bottleneck has also been analyzed explicitly (Wang & Palmer, 2023). Other works impose related geometry directly: orthogonal classifier weights can improve adversarial robustness (Xu et al., 2022); ETF classifiers have been used for early exit (Ji et al., 2023); and fixed or hierarchy-aware frames have been used to reduce classifier bias or mistake severity (Tong & Davis, 2023). Neural Collapse by Design trains features and prototypes jointly on the unit hypersphere so that the global optimum of the NONL contrastive objective is exactly the simplex ETF (Koromilas et al., 2026); we include it as a classification-only comparison baseline. Equidistant prototype embeddings provide an earlier prototype-based classification example (Deng et al., 2014).

These works constrain features, classifier weights, or prototypes through a classification loss or a fixed readout. OPB addresses a different object: the label-conditional Gaussian prior in the bottleneck. Its polar/QR frame is recomputed from the prior encoder, and the anchor-energy classifier reads the same frame without introducing an independent classifier scale. Thus the geometric class code and the conditional KL are coupled during training.

Distance-preserving regression and geometric priors. Isometric representation learning has traditionally focused on preserving distances on a data manifold, including deep extensions of isomap-style mappings (Pai et al., 2022). Orthogonal label regression has also been used as a discriminative device in domain adaptation (Luo et al., 2023). These methods motivate using a declared label metric, but they do not formulate that metric as the mean map of a conditional Gaussian information-bottleneck prior. EPB uses the scalar label case: after standardization, prior means lie on a normalized axis with a fixed scale, so latent prior distances equal the declared label distances up to that scale. The regression information-bottleneck literature, including divergence-based bounds, provides complementary compression objectives rather than this geometric prior construction (Yu et al., 2024). AdaCap pairs a permutation-based contrastive loss with a Tikhonov closed-form output layer for small-sample tabular regression (Belucci et al., 2026); we include it as a regression-only comparison baseline.

Geometry-aware information bottlenecks. Recent work has used the phrase geometric information bottleneck in several distinct settings. A variational recommender learns a geometric prior for explanation generation (Yan et al., 2024); a data-scarcity formulation penalizes latent curvature and intrinsic dimension (Katende, 2025); and activation steering uses orthogonal subspaces to control language-model interventions (Doan et al., 2026). A concurrent analysis studies the relationship between information compression and class-wise geometric compression in CEB networks (Adilova et al., 2026). These studies make geometry an important representation concern, but they do not use the same conditional Gaussian prior for both compression and prediction, nor do they provide the OPB/EPB alignment-to- separation construction considered here.

AI USE STATEMENT

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools for drafting and editing portions of the code and of this manuscript. All AI-assisted content was manually reviewed and verified by the authors; experimental results and theoretical claims were produced and checked by our own code and analysis. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The code, hyperparameter grids, and evaluation protocol are described in Section 3.1; the geometric prior is stateless (no EMA or prototypes) and the orthogonalization is exact wherever defined (Assumption 2.1). All datasets are public, with preprocessing steps given in the release. Proofs of all statements are collected in the appendix. Statistical claims use the paired bootstrap confidence intervals and non-inferiority margin described in Appendix A.6.

Acknowledgments

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A Deferred proofs, construction details, and scope

A.1 Failure diagnosis

Proposition A.1 (Dead knob, following Caveat 1 of Kolchinsky & Tracey (2019))**.** *Assume $Y = f(X)$. For every $\gamma > 0$, the CEB objective $L_\gamma(Z) := I(X;Z|Y) - \gamma I(Y;Z)$ is minimized exactly on the corner family $\{Z : I(Y;Z) = H(Y),\ I(X;Z|Y) = 0\}$, and the map $\gamma \mapsto \big(I(X;Z^*|Y),\, I(Y;Z^*)\big)$ is constant. Consequently no interior point of the IB curve is ever a Lagrangian minimizer of CEB, over the entire parameter range.*

Proof sketch (see Kolchinsky & Tracey (2019) for the full argument). By the chain rule, $L_\gamma(Z) = I(X;Z) - (1+\gamma)I(Y;Z)$; the data-processing inequality gives $L_\gamma \geq -\gamma H(Y)$, attained exactly at the corner $Z = Y$. In the Caveats paper's maximizing convention this is their Eq. 7 with $\beta' = 1/(1+\gamma) \in (0,1)$, the range in which every maximizer is pinned at the copy corner. ☐

The corner is a single point of the information plane and an entire equivalence class of representations; the objective is exactly indifferent within the class, and neither the objective nor the certificate records which member the optimizer reaches. In the deterministic scenario this corner *is* the MNI point, so CEB's declared target and the degenerate attractor coincide. The Caveats paper repairs this by squaring the compression term; our anchored reference supplies the same strict convexity by a different mechanism, without squaring: a quadratic displacement term built into the KL by construction (Proposition A.4 below). We present this as a mechanism connection to the Caveats fix, not as a new solution of the deterministic degeneration.

Proof of Proposition 2.1. For any matched pair ($b = q(z|y)$), the gap identity of Eq. equation 1 gives Eq. equation 2; the matching condition is stated per class and no term of the certificate couples two distinct class priors, so the certificate value carries no information about the pairwise geometry of $\{q(z|y)\}$. Note that $I(X; Z|Y)$ is generally non-zero; it vanishes only under the conditional independence $Z \perp X \mid Y$, a property of the encoder, not of the prior. □

Proposition A.2 (Retention blindness on T_α). *On the manifold $T_\alpha = B_\alpha \cdot Y$ of Kolchinsky & Tracey (2019) (their Eq. 6), a family that depends on Y alone, $I(X; T_\alpha|Y) = 0$ for every α, and with the optimal backward encoder and classifier,* $\mathrm{ReX}(T_\alpha) = 0$ *and* $\mathrm{CE}(T_\alpha) = H(Y) - I(Y; T_\alpha) =: \delta_\alpha$*: the certificate reads 0 from the total collapse ($\alpha = 0$) to the full copy ($\alpha = 1$), while the retained label information $I(Y; T_\alpha)$ sweeps the whole axis. Tightness of the certificate is therefore compatible with retaining all, part, or none of Y: the certificate records nothing about how much label information the code retains.*

Proof. T_α is a function of Y alone, hence $T_\alpha \perp X \mid Y$ and $I(X; T_\alpha|Y) = 0$. By Proposition 2.1 with the true backward encoder the certificate is tight, $\mathrm{ReX} = I(X; T_\alpha|Y) = 0$, and the optimal classifier attains $\langle \log c(y|z) \rangle = -H(Y|T_\alpha)$, i.e. $\mathrm{CE} = H(Y|T_\alpha) = \delta_\alpha$. □

Remark A.1 (The objective on T_α is β-independent and strictly prefers the full copy). On T_α the compression term contributes nothing ($\mathrm{ReX} \equiv 0$), so the objective reads $L = \mathrm{CE} + \beta\,\mathrm{ReX} = \delta_\alpha$, independent of β and strictly minimized at $\alpha = 1$ (the full copy). No certified residual is produced anywhere on the manifold; the blindness is that the certificate cannot tell the optimizer — or the analyst — where on the retention axis the code sits.

Scope of the deterministic comparison-solution argument. In Theorem 2.1 the comparison solution $q(z|x) = p_\theta(z|y)$ requires a zero-KL feasible solution. The condition $Y = f(X)$ alone does not provide this feasibility: the chosen encoder must recover the label at the required level and the posterior family must represent the prior covariance, while the prediction head must represent the corresponding aligned predictor. These are realizability assumptions on the data/model pair, not properties established for every benchmark dataset. For tasks in which any of them fails (including realistic settings with label ambiguity, finite model capacity, or a restricted head), the $O(1/\beta)$ bound on $\mathbb{E}[D(Y)]$ is not available. In particular, for continuous regression where the encoder cannot recover y from x and no such Bayes comparison classifier is defined, part (a) does not apply: there the separation transfer of Proposition 2.4 remains a conditional statement on the anchor mismatch, and no framework-level alignment bound is claimed. Part (b) is a separate deterministic-regression argument with its own comparison solution. The benchmark experiments therefore test the realized mismatch and posterior geometry directly; they do not assume or verify all premises of Theorem 2.1.

A.2 Proofs of the conditional guarantees

Proof of Proposition 2.3. For fixed θ, apply the gap identity of Eq. equation 1 to the conditional distribution $p_\theta(z|y)$ and the posterior q_ϕ. □

Proof of Proposition 2.4. For (a), Jensen gives $\|m_k - m_\theta(k)\| \le \sqrt{2\tau_{\max}^2 D_k}$, and the triangle inequality on $m_i - m_j = (m_i - m_\theta(i)) + (m_\theta(i) - m_\theta(j)) + (m_\theta(j) - m_j)$ gives Eq. equation 16. For (b), the same two inequalities hold pointwise for P_Y-a.e. y: Jensen on the regular conditional expectation gives $\|m(y) - m_\theta(y)\| \le \sqrt{2\tau_{\max}^2 D(y)}$ (the norm is convex), and the triangle inequality gives Eq. equation 18; intersecting the two measure-one sets of valid y's yields the almost-everywhere pair statement. □

Remark A.2 (Finite-sample estimation of the continuous case). On finite samples the regular conditional expectations of part (b) are estimated by binning the label axis into intervals B_i with representative labels $\bar{y}_i$ and centers $m_i = \mathbb{E}[\mu_\phi(X) \mid Y \in B_i]$; the bin diameter adds a quantization error bounded by $\rho_{\max}\, \mathrm{diam}(B_i) + \rho_{\max}\, \mathrm{diam}(B_j)$ (by the upper bound of Eq. equation 5), which vanishes as the bins shrink. The theorem itself is stated pointwise and needs no binning.

Proof of Theorem 2.1. *(a)* Since $\Sigma_\theta(y) \preceq \tau_{\max}^2 I_d$, each sample's KL contains a mean-mismatch term with variance at most $\tau_{\max}^2$, so $\mathrm{KL}(q_\phi(z|x) \,\|\, p_\theta(z|y)) \geq \|\mu_\phi(x) - m_\theta(y)\|^2/(2\tau_{\max}^2)$, and taking expectations gives $\mathbb{E}[\mathrm{KL}] \geq \mathbb{E}[D(Y)]$. The comparison solution $q(z|x) = p_\theta(z|y)$ is feasible by the assumptions ($Y = f(X)$, class-constant maps, diagonal representable covariances), has zero KL, and its prediction is the Bayes posterior of Eq. equation 19, whose cross-entropy is exactly the conditional entropy $C_{\mathrm{G}} = H(Y \mid Z_{\mathrm{align}}) \leq \ln K$; for the orthogonal instance the anchor-energy classifier of the main model realizes it (shared covariance, determinant factors cancel). Hence $L^* \leq C_{\mathrm{G}}$ and $\beta\,\mathbb{E}[\mathrm{KL}] \leq L \leq L^* + \varepsilon \leq C_{\mathrm{G}} + \varepsilon \leq \ln K + \varepsilon$, whence Eq. equation 20.

(b) As in (a), $\mathbb{E}[\mathrm{KL}] \geq \mathbb{E}[D(Y)]$ by the covariance bound. The comparison solution is feasible by the assumptions ($Y = f(X)$, aligned maps representable): its posterior is $q(z|x) = p_\theta(z|y)$ (zero KL) and its prediction is the tied head, whose risk is $C_{\mathrm{align}} = \tau^2/\rho^2$ (the prediction error is exactly the Gaussian bottleneck noise $(\tau/\rho)\,\eta$). Hence $L^* \leq C_{\mathrm{align}}$ and $\beta\,\mathbb{E}[\mathrm{KL}] \leq L \leq L^* + \varepsilon \leq C_{\mathrm{align}} + \varepsilon$, whence Eq. equation 21. □

Proof of Corollary 2.1. *(a)* $D_k \leq \mathbb{E}[D(Y)]/\pi_k \leq (C_{\mathrm{G}} + \varepsilon)/(\beta\pi_k)$; combine with Proposition 2.4, whose mismatch terms are $\sqrt{2\tau_{\max}^2 D_k}$. *(b)* Markov's inequality on the non-negative variable $D(Y)$ with mean at most $(C_{\mathrm{align}} + \varepsilon)/\beta$ gives $P_Y(D(Y) > t) \leq (C_{\mathrm{align}} + \varepsilon)/(\beta t)$; on S_t the pointwise transfer of Proposition 2.4(b) applies with $D(y), D(y') \leq t$. □

Proof of Proposition 2.5. By Jensen's inequality the left side is at least $\frac{1}{2\tau^2}\sum_k \pi_k \|c - a q_{\theta,k}\|^2 = \frac{1}{2\tau^2}\big(\|c\|^2 - 2a\, c^\top \sum_k \pi_k q_{\theta,k} + a^2\big)$. Minimizing over c gives $c = a\sum_k \pi_k q_{\theta,k}$; by orthonormality $\|\sum_k \pi_k q_{\theta,k}\|^2 = \sum_k \pi_k^2$, yielding Eq. equation 24. □

Remark A.3 (What the framework constrains (and what it does not)). The constraints of the geometric prior bind the reference, not the deployed representation: for any orthonormal frame $\{q'_k\}$ and the class-constant posterior $q(z|Y = k) = \mathcal{N}(a\,q'_k, \tau^2 I_d)$, the trainable frame can rotate to match $(q_{\theta,k} \to q'_k)$, yielding $\mathrm{KL} = 0$, $I(X;Z|Y) = 0$, and $I(Y;Z) > 0$: a full-retention label code with a zero certificate. The framework therefore constrains the geometric *form* of the label code (precisely what CEB left unspecified), and does not claim that the certificate meters retained label information. (With the variance fixed, the remaining freedom at certificate zero is the frame rotation; a learnable per-class variance would add a further circumvention channel and is deliberately not used.)

A.3 Construction details

Polar permutation equivariance. For any label permutation matrix P,

$$Q_\theta(U_\theta P) = U_\theta P\big(P^\top U_\theta^\top U_\theta P\big)^{-1/2} = U_\theta\big(U_\theta^\top U_\theta\big)^{-1/2} P = Q_\theta(U_\theta)\,P, \tag{26}$$

since $PP^\top = I_K$ and the inverse square root conjugates under P. Permuting the label numbering therefore permutes the anchors accordingly, and no class index plays a distinguished role.

Proposition A.3 (Orthogonal class-prior geometry by construction)**.** *Under Eqs. equation 8–equation 9, the class means $a_k = a\,q_{\theta,k}$ satisfy Eq. equation 10; equivalently, the class-mean Gram matrix is $a^2 I_K$: under Assumption 2.1, the prior supplies a class-separated spherical code at every training step, whatever the optimizer does.*

Proof. Eq. equation 10 follows from $Q_\theta^\top Q_\theta = I_K$ after scaling by a; in particular $\|a_i - a_j\|^2 = \|a_i\|^2 + \|a_j\|^2 - 2a_i^\top a_j = 2a^2$. □

Initialization, monitoring, and re-initialization. The posterior logvar head is zero-initialized ($\sigma^2 = \tau^2 = 1$ at initialization, so training starts at $\mathrm{KL} \approx \frac{1}{2\tau^2}a^2$); the prior encoder is identity-initialized, so the initial raw matrix is $U_\theta = [e_1, \dots, e_K]$, full column rank, with a stable polar-factor backward (the verified condition of Assumption 2.1). The full-rank condition is *verified, not imposed*: $\sigma_{\min}(U_\theta)$ is monitored along training and stays bounded away from zero in the reported runs, and no additional regularizer is added to the objective, which remains Eq. equation 11 exactly. Near rank deficiency the polar factor's gradient is delicate; should the monitor flag a near-deficient step, the recommended handling is a re-initialization of the prior encoder (the reported runs never triggered it).

EPB axis choices. Three variants of the direction: (A) *fixed axis*, $u = [1, 0, \ldots, 0]$, with no trainable prior direction: the theoretically cleanest ablation, at the price of prespecifying a latent coordinate; (B) *trainable normalized axis*, $u = W/\|W\|$ with W trained and normalized at every step, our main variant, which keeps the scale ρ exact while allowing the model to choose the axis; (C) *unconstrained linear axis*, $\mu_p(y) = W\tilde{y}$: the simplest ablation, still isometric by linearity but with a free scale that can shrink and weaken the geometry. We recommend (B), use (A) to test whether the trainable rotation carries the benefit, and keep (C) as a diagnostic. The main variant uses no bias: an affine offset cancels in pairwise distances but is a free extra degree of matching, and $\tilde{y}$ is already centered.

EPB isometry notes. The isometry is exact and global: any nonzero linear map of a scalar label is isometric up to its norm, and the normalization fixes that norm to ρ; unlike fixed random-feature anchors, whose pairwise distances grow nonlinearly and saturate, the isometric axis preserves label distances at every scale (with the price that anchor norms grow with $|\tilde{y}|$; a line and a fixed-radius sphere intersect in at most two points, so isometry and equal norms cannot both hold). All prior means lie on one line; for $y_i < y_j < y_k$ the distances add, $\|\mu_p(y_i) - \mu_p(y_k)\| = \|\mu_p(y_i) - \mu_p(y_j)\| + \|\mu_p(y_j) - \mu_p(y_k)\|$: order and magnitude of label differences are encoded exactly.

Multidimensional regression labels. For $y \in \mathbb{R}^m$ ($m \le d$), the same construction extends verbatim: take $W \in \mathbb{R}^{d\times m}$, its thin polar factor $Q = W(W^\top W)^{-1/2}$, and the prior $\mu_p(y) = \rho\, Q\, \tilde{y}$, which is isometric in the label metric; the scalar case is $m = 1$ with $Q = u$. The label metric must be declared before standardization, otherwise "isometric" has no unique meaning.

A.4 Additional results

Lemma A.1 (OPB KL decomposition)**.** *For a diagonal posterior covariance and the prior of Eq. equation 9,*

$$\mathrm{KL}\big(q_\phi(z|x)\,\|\,p_\theta^{\mathrm{OPB}}(z|y)\big) = \frac{1}{2\tau^2}\|\mu_\phi(x) - a\,q_{\theta,y}\|^2 + \frac{1}{2}\sum_{r=1}^{d}\Big[\frac{\sigma^2_{\phi,r}(x)}{\tau^2} - 1 - \log\frac{\sigma^2_{\phi,r}(x)}{\tau^2}\Big], \quad (27)$$

a class-anchor mismatch term plus a variance mismatch term that is non-negative and vanishes exactly at $\sigma^2_{\phi,r}(x) = \tau^2$ for all r. The mismatch term is quadratic in the displacement to a reference whose scale and covariance cannot move: this is the mechanism behind the strict convexity of Proposition A.4 below.

Proof. Apply the diagonal-Gaussian KL formula with posterior mean $\mu_\phi(x)$, prior mean $a\,q_{\theta,y}$, posterior variances $\sigma^2_{\phi,r}(x)$, and the fixed prior variance τ^2. □

Corollary A.1 (Label-information lower bound in the aligned Gaussian model)**.** *Assume uniform classes and exact alignment $q(z|Y = k) = \mathcal{N}(a\,q_{\theta,k}, \tau^2 I_d)$. The pairwise error probability of the nearest-anchor classifier is $P_{\mathrm{pair}} = \Phi\big(-\frac{a}{\sqrt{2}\,\tau}\big)$, with $P_e \le \min\{1, (K-1)P_{\mathrm{pair}}\}$, and whenever the displayed bound $\bar{P}_e \le 1 - 1/K$, Fano's inequality gives $I(Y;Z) \ge \log K - h_2(\bar{P}_e) - \bar{P}_e\log(K-1)$. This links the anchor scale-to-noise ratio a/τ to the retained label information, conditionally on the aligned Gaussian model.*

Proof. Two anchors have distance $\sqrt{2}a$ by Proposition A.3; the equal-covariance binary decision boundary is halfway between them, giving $\Phi(-\sqrt{2}a/(2\tau))$. The union bound and Fano's inequality on $[0, 1 - 1/K]$ conclude. □

Proposition A.4 (Joint strict convexity and unique optimum on a restricted aligned surrogate)**.** *Consider the class-symmetric aligned family: every input of class y is mapped to the posterior $q_y = \mathcal{N}(s\,q_{\theta,y}, \sigma^2 I)$ ($s \ge 0$, $\sigma^2 > 0$; anchor scale $a > 0$), and the classifier is the anchor-energy classifier of Proposition 2.2 (weights $w_j = c\,q_{\theta,j}$ with scale $c = a/\tau^2$ fixed by construction). With the* stochastic training code *$z \sim q_y$ (as in the actual objective, Eq. equation 11), the training cross-entropy is*

$$g(s, \sigma^2) := \mathbb{E}_\eta\Big[\ln\Big(1 + \textstyle\sum_{j\neq y} e^{-cs + c\sigma(\eta_j - \eta_y)}\Big)\Big], \qquad \eta \sim \mathcal{N}(0, I_K), \quad (28)$$

and the training Lagrangian is

$$L_\beta(s,\sigma^2) = g(s,\sigma^2) + \beta\Big[\tfrac{1}{2\tau^2}(s-a)^2 + \tfrac{d}{2}\big(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)\big)\Big]. \tag{29}$$

Then for every $\beta > 0$, L_β is strictly convex jointly in (s,σ) and has a unique joint minimizer $(s^*(\beta),\sigma^*(\beta))$, interior in s since $a > 0$: the stochastic training objective has a well-defined optimum, in contrast to the deterministic IB Lagrangian, whose optimum is pinned at a corner for every β (Proposition A.1). This is a restricted surrogate result*: it analyzes two scalar variables on the class-symmetric aligned family, with the frame, the anchor scale, and the classifier scale held fixed; the full OPB objective, with its nonlinear posterior, trainable frame, and non-convex degrees of freedom, lies outside this analysis, and the result neither establishes β-tunability of the full objective nor makes any claim on the information plane. The deterministic evaluation code ($z = \mu$) is the $\sigma \to 0$ limit, and $g \geq \ln(1 + (K-1)e^{-cs})$ by Jensen (the log-sum-exp is convex), so the statement strictly generalizes the deterministic surrogate analysis. We do* not *claim here that the map $\beta \mapsto s^*(\beta)$ is monotone or that the sweep is one-to-one: the minimizer σ^* responds to β through the cross-derivative $\partial^2 g/\partial s\,\partial\sigma$, and establishing the sign of $\mathrm{d}s^*/\mathrm{d}\beta$ requires the full two-dimensional stationarity analysis, which we leave open; the β-sweep is reported empirically in the experiments. The fixed-scale condition holds for the main model by construction (Proposition 2.2), so the statement describes the actual training objective up to the aligned-family idealization. We do not analyze the free-classifier variant on this surrogate: under the stochastic training code, a diverging classifier scale does not produce a degenerate solution: at $\mathrm{KL} \to 0$ the posterior variance is pinned at $\sigma^2 = \tau^2 > 0$, so the class-conditional Gaussians overlap and the training cross-entropy of a free classifier diverges with its scale rather than vanishing (for $K = 2$ it contains $\mathbb{E}[\ln(1 + e^{-c(a+\tau\xi)})]$, whose integrand grows linearly in c on the event $\xi < -a/\tau$, which has positive probability). An analysis of the free classifier would require an explicit classifier-norm constraint or regularizer and is left to future work.*

Proof. For each fixed η, the integrand of Eq. equation 28 is a log-sum-exp of functions affine in (s,σ) (the constant 1 has slope zero and each exponential has slope $(-c,\ c(\eta_j - \eta_y))$), hence jointly convex in (s,σ), and the expectation g is jointly convex. No strictness is claimed for g: for $K = 2$ the integrand depends on (s,σ) only through the single linear combination $-cs + c\sigma(\eta_2 - \eta_1)$, so its Hessian has rank one. Write the KL term as $R(s,\sigma) := \frac{1}{2\tau^2}(s-a)^2 + \frac{d}{2}\big(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)\big)$; its Hessian in (s,σ) is $\nabla^2 R = \mathrm{diag}\big(\tau^{-2},\ d\,(\tau^{-2} + \sigma^{-2})\big)$, strictly positive definite for every $\sigma > 0$, so βR is strictly convex in (s,σ). Hence $L_\beta = g + \beta R$ is the sum of a jointly convex and a strictly convex function, and is strictly convex in (s,σ). Since $R \to +\infty$ as $s \to \infty$, $\sigma \to 0^+$, or $\sigma \to \infty$, the sublevel sets of L_β are compact, and the unique joint minimizer exists (by strict convexity it is unique). For $a > 0$ the minimizer is interior in s: at $s = 0$, $\partial L_\beta/\partial s = -c\,\mathbb{E}[p_{\mathrm{w}}] - \beta a/\tau^2 < 0$, where p_{w} is the error probability of the sampled code, so $s^* > 0$. □

Remark A.4 (The sweep is a KL–CE surrogate curve; scope and limitations). Proposition A.4 proves joint strict convexity and uniqueness of the optimum for the stochastic training objective in the KL–CE plane on the aligned family at fixed classifier scale: a restricted surrogate result, which does not establish β-tunability of the full OPB objective, and in any case not a statement on the information plane of the Caveats paper; the results of the main text establish a structured conditional prior, an exact KL geometry, and conditional separation guarantees, and they do *not* establish a mutual information upper bound tighter than CEB's: Eq. equation 14 is the usual CEB gap identity for a particular prior family. The orthogonal-frame construction fixes all pairwise anchor distances, so the framework should not claim to recover arbitrary semantic class similarities. All geometric statements are conditional on Assumption 2.1: orthonormality holds exactly by the polar construction wherever the factorization exists, and the condition is verified, not imposed, along the reported runs (Section 2.2.1); the construction is exactly equivariant under relabeling of the classes by Eq. equation 26, so no class index plays a distinguished role. If the frame were computed from the current input batch rather than from the all-class prior matrix, the prior would become batch-dependent and Proposition 2.3 would no longer follow directly. Finally, the anchor-energy classifier of Proposition 2.2 fixes the classifier scale to a/τ^2 by construction, removing the independent classifier-margin escape route in the aligned surrogate of Proposition A.4; whether a free classifier (with an explicit norm constraint or regularizer) admits its own degenerate route is left to future work, and the free linear classifier is kept in the experiments as a variant for comparison. Whether the joint optimum of Proposition A.4

Table 7: Test accuracy (%) and test R^2 with standard deviations over runs: the plain backbone, the variational baselines, and NCBD (classification settings only); same configurations as Table 2.

Dataset	Backbone	Base	VIB	SVIB	NIB	NCBD
MNIST	MLP	98.33±0.12	98.28±0.11	98.15±0.13	98.27±0.07	97.95±0.23
MNIST	CNN	99.13±0.05	99.28±0.09	99.23±0.03	99.22±0.06	99.18±0.07
ImageNet-100	MLP	83.12±0.14	82.84±0.44	83.02±0.37	82.35±0.27	83.64±0.07
ImageNet-100	CNN	83.40±0.13	82.94±0.37	83.11±0.86	82.91±0.35	80.56±1.86
Cora	GCN	87.68±0.46	86.83±0.28	86.46±1.87	86.35±0.68	87.34±1.60
IMDb	RNN	84.85±0.68	85.50±0.91	85.99±0.30	86.14±0.92	86.69±0.52
AG News	RNN	**92.94±0.15**	92.72±0.17	92.86±0.21	92.91±0.22	92.37±0.37
Cal. Housing	MLP	0.7618±0.0238	<u>0.7901±0.0049</u>	0.7897±0.0058	0.7862±0.0035	–
STS-B	RNN	**0.2899±0.0113**	0.2719±0.0164	0.2697±0.0202	0.2417±0.0221	–
ZINC	GCN	<u>0.6348±0.0044</u>	0.6339±0.0033	0.6325±0.0026	0.6178±0.0099	–
AgeDB	MLP	0.2180±0.0409	0.3173±0.0101	0.3385±0.0286	0.2710±0.0235	–
AgeDB	CNN	0.3914±0.0194	0.3342±0.1894	0.3323±0.1870	0.3337±0.1889	–

Table 8: Test accuracy (%) and test R^2 with standard deviations over runs: CEB, DVCCA, AdaCap (regression settings only), and our method; same configurations as Table 2. GPB and GPB-L are separated from the references by the vertical rule.

Dataset	Backbone	CEB	DVCCA	AdaCap	GPB	GPB-L
MNIST	MLP	<u>98.43±0.08</u>	98.24±0.17	–	**98.54±0.09**	<u>98.43±0.07</u>
MNIST	CNN	<u>99.30±0.05</u>	99.29±0.10	–	**99.32±0.05**	<u>99.31±0.05</u>
ImageNet-100	MLP	82.77±0.22	82.99±0.43	–	**84.40±0.21**	<u>84.01±0.23</u>
ImageNet-100	CNN	82.97±0.67	83.21±0.70	–	**86.08±0.21**	<u>84.88±0.80</u>
Cora	GCN	86.27±0.38	86.90±0.94	–	**88.12±0.55**	<u>87.86±0.87</u>
IMDb	RNN	85.55±0.73	85.69±0.23	–	<u>86.84±0.57</u>	**86.99±0.35**
AG News	RNN	92.83±0.11	92.78±0.15	–	<u>92.87±0.12</u>	<u>92.92±0.14</u>
Cal. Housing	MLP	0.7894±0.0018	0.7878±0.0048	0.7758±0.0023	0.7895±0.0051	**0.7914±0.0034**
STS-B	RNN	0.2668±0.0127	0.2713±0.0218	0.1613±0.0188	<u>0.2762±0.0152</u>	0.2755±0.0134
ZINC	GCN	0.6257±0.0044	0.6294±0.0026	**0.6770±0.0019**	0.6344±0.0031	0.6323±0.0016
AgeDB	MLP	0.3317±0.0251	0.3180±0.0228	<u>0.3482±0.0209</u>	**0.3536±0.0076**	0.3320±0.0080
AgeDB	CNN	0.3882±0.0275	0.3205±0.1811	<u>0.4684±0.0094</u>	**0.4700±0.0072**	0.4388±0.0220

moves monotonically in β is likewise open: it requires the full two-dimensional stationarity analysis, in which the optimal variance responds to β through the cross-derivative $\partial^2 g/\partial s\,\partial\sigma$ of the stochastic cross-entropy. The trainable frame and axis constitute a rotation channel that this paper does not address: the certificate's zero set contains the full-retention orthogonal label codes at any frame rotation (the reference can rotate to meet them), and the framework pins the *geometry of the code*; it does not claim that the certificate reads off the retained label information. Quantifying how much of the anchor gap is absorbed by frame rotation rather than by posterior motion is left to future work (a frozen-frame ablation isolates it).

A.5 Main results with standard deviations

Tables 7 and 8 report the same configurations as the main table with standard deviations over runs, split by column groups for width (the baselines table includes NCBD and the variants table includes AdaCap); bold and underline follow the same mean-based rule as Table 2.

A.6 Paired-difference confidence intervals

Tables 9–12 report, for each setting, the paired difference of GPB over every baseline (GPB minus baseline) with a bootstrap 95% confidence interval: the differences are paired across the shared seeds, the interval is the percentile interval of $B = 10{,}000$ resamples of the per-run differences (fixed seed, reproducible), and the non-inferiority margin is $\delta = 0.2$ percentage points of accuracy or $\delta = 0.005$ in R^2. Bold marks a lower bound above zero (significant advantage) and † a lower bound within the margin (tied); the best configuration per method is the same as in the main table. The last table covers the external non-IB baselines, NCBD on the classification settings and AdaCap

Table 9: Paired-difference 95% confidence intervals of GPB over the plain backbone and VIB (GPB minus baseline, per setting; classification in percentage points, regression in R^2). Bold: lower CI bound above zero; †: lower bound within the non-inferiority margin δ (tied).

Dataset	Backbone	Base	VIB
MNIST	MLP	**+0.2 [+0.2, +0.3]**	**+0.3 [+0.2, +0.3]**
MNIST	CNN	**+0.2 [+0.2, +0.2]**	+0.0 [-0.0, +0.1]†
ImageNet-100	MLP	**+1.3 [+1.2, +1.4]**	**+1.6 [+1.4, +1.9]**
ImageNet-100	CNN	**+2.7 [+2.5, +2.9]**	**+3.1 [+3.0, +3.4]**
Cora	GCN	+0.4 [-0.1, +1.1]†	**+1.3 [+0.7, +1.7]**
IMDb	RNN	**+2.0 [+1.2, +2.7]**	**+1.3 [+0.6, +2.3]**
AG News	RNN	-0.1 [-0.2, +0.0]†	**+0.2 [+0.1, +0.2]**
Cal. Housing	MLP	**+0.028 [+0.008, +0.048]**	-0.001 [-0.005, +0.003]†
STS-B	RNN	-0.014 [-0.022, -0.006]	+0.004 [-0.017, +0.022]
ZINC	GCN	-0.000 [-0.003, +0.002]†	+0.001 [-0.004, +0.004]†
AgeDB	MLP	**+0.136 [+0.104, +0.167]**	**+0.036 [+0.026, +0.046]**
AgeDB	CNN	**+0.079 [+0.066, +0.091]**	**+0.136 [+0.043, +0.302]**
		8*/3†	8*/3†

Table 10: Paired-difference 95% confidence intervals of GPB over SVIB and NIB (GPB minus baseline, per setting; classification in percentage points, regression in R^2). Bold/† as in the baselines table.

Dataset	Backbone	SVIB	NIB
MNIST	MLP	**+0.4 [+0.3, +0.5]**	**+0.3 [+0.2, +0.3]**
MNIST	CNN	**+0.1 [+0.1, +0.1]**	**+0.1 [+0.1, +0.2]**
ImageNet-100	MLP	**+1.4 [+1.1, +1.7]**	**+2.1 [+1.8, +2.2]**
ImageNet-100	CNN	**+3.0 [+2.4, +3.9]**	**+3.2 [+2.9, +3.5]**
Cora	GCN	**+1.7 [+0.2, +3.4]**	**+1.8 [+1.1, +2.4]**
IMDb	RNN	**+0.8 [+0.2, +1.4]**	**+0.7 [+0.2, +1.2]**
AG News	RNN	+0.0 [-0.1, +0.1]†	-0.0 [-0.2, +0.1]†
Cal. Housing	MLP	-0.000 [-0.006, +0.004]	+0.003 [-0.001, +0.009]†
STS-B	RNN	+0.006 [-0.021, +0.028]	**+0.035 [+0.011, +0.058]**
ZINC	GCN	+0.002 [-0.001, +0.004]†	**+0.017 [+0.010, +0.026]**
AgeDB	MLP	+0.015 [-0.010, +0.041]	**+0.083 [+0.062, +0.100]**
AgeDB	CNN	**+0.138 [+0.045, +0.303]**	**+0.136 [+0.040, +0.302]**
		7*/2†	10*/2†

on the regression settings; the only interval lying entirely below zero is AdaCap on ZINC (-0.043 R^2).

A.7 Information-plane panels for Cal. Housing

Figure 8 completes Fig. 3 of the main text with the regression setting, under the same β grid, axis conventions, and five runs per point (Section 3.6); here $I(Y;Z)$ is the Gaussian approximation $\frac{1}{2}\ln\big(\mathrm{Var}(y)/\mathrm{Var}(y-\hat{y})\big)$, a heuristic rather than a bound. On Cal. Housing this approximation consistently reads above the InfoNCE bound, so the certificate panel reflects the estimator gap rather than the geometry; the information plane remains informative: CEB compresses itself off the task ($R^2 = 0.49$ at $\beta = 0.1$, 0.18 at $\beta = 0.5$) while every EPB curve keeps $R^2 \geq 0.76$ flat out to $\beta = 25$.

Table 11: Paired-difference 95% confidence intervals of GPB over CEB, DVCCA, and the free-head ablation GPB-L (GPB minus baseline, per setting; classification in percentage points, regression in R^2). Bold/† as in the baselines table.

Dataset	Backbone	CEB	DVCCA	GPB-L
MNIST	MLP	**+0.1 [+0.0, +0.2]**	**+0.3 [+0.2, +0.4]**	**+0.1 [+0.0, +0.2]**
MNIST	CNN	+0.0 [-0.0, +0.1]†	+0.0 [-0.1, +0.1]†	+0.0 [-0.0, +0.1]†
ImageNet-100	MLP	**+1.6 [+1.5, +1.7]**	**+1.4 [+1.1, +1.7]**	**+0.4 [+0.2, +0.5]**
ImageNet-100	CNN	**+3.1 [+2.4, +3.7]**	**+2.9 [+2.4, +3.5]**	**+1.2 [+0.5, +1.9]**
Cora	GCN	**+1.8 [+1.5, +2.3]**	**+1.2 [+0.4, +1.9]**	+0.3 [-0.5, +1.3]
IMDb	RNN	**+1.3 [+0.3, +2.1]**	**+1.1 [+0.8, +1.5]**	-0.2 [-0.6, +0.5]
AG News	RNN	+0.0 [-0.1, +0.2]†	+0.1 [-0.0, +0.2]†	-0.0 [-0.2, +0.1]†
Cal. Housing	MLP	+0.000 [-0.003, +0.004]†	+0.002 [-0.005, +0.009]	-0.002 [-0.005, +0.002]
STS-B	RNN	+0.009 [-0.009, +0.025]	+0.005 [-0.020, +0.026]	+0.001 [-0.019, +0.017]
ZINC	GCN	**+0.009 [+0.005, +0.011]**	**+0.005 [+0.002, +0.007]**	+0.002 [-0.001, +0.005]†
AgeDB	MLP	**+0.022 [+0.006, +0.038]**	**+0.036 [+0.012, +0.057]**	**+0.022 [+0.019, +0.025]**
AgeDB	CNN	**+0.082 [+0.064, +0.099]**	**+0.149 [+0.065, +0.310]**	**+0.031 [+0.018, +0.046]**
		8*/3†	8*/2†	5*/3†

Table 12: Paired-difference 95% confidence intervals of GPB over the external non-IB baselines NCBD (classification settings only) and AdaCap (regression settings only); GPB minus baseline, classification in percentage points, regression in R^2. Bold/† as in the baselines table.

Dataset	Backbone	NCBD	AdaCap
MNIST	MLP	**+0.6 [+0.4, +0.8]**	–
MNIST	CNN	**+0.1 [+0.1, +0.2]**	–
ImageNet-100	MLP	**+0.8 [+0.6, +0.9]**	–
ImageNet-100	CNN	**+5.5 [+4.3, +6.9]**	–
Cora	GCN	**+0.8 [+0.0, +1.9]**	–
IMDb	RNN	+0.2 [-0.5, +0.7]	–
AG News	RNN	**+0.5 [+0.2, +0.8]**	–
Cal. Housing	MLP	–	**+0.014 [+0.009, +0.019]**
STS-B	RNN	–	**+0.115 [+0.086, +0.135]**
ZINC	GCN	–	-0.043 [-0.046, -0.041]
AgeDB	MLP	–	+0.005 [-0.007, +0.021]
AgeDB	CNN	–	+0.002 [-0.004, +0.006]†
		6*/0†	2*/1†

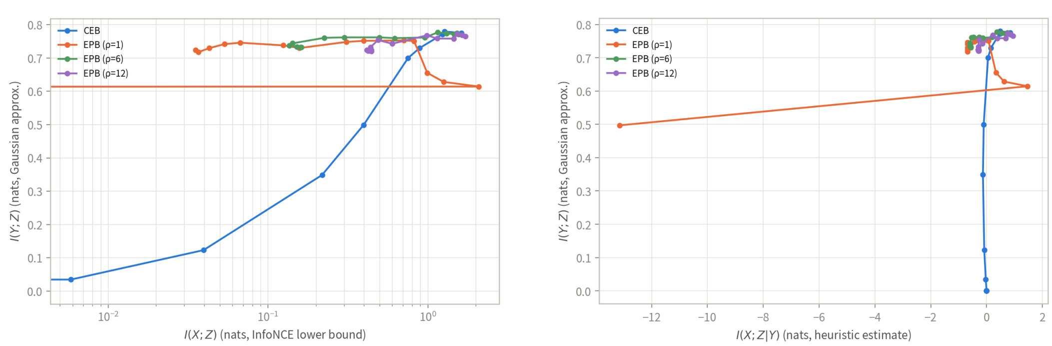

Figure 8: Information plane (a) and certificate plane (b) for Cal. Housing, companion panels to Fig. 3: identical estimators, β grid (log x-scale in (a); linear where the estimated residual turns negative in (b)), five runs per point, lines connect β in ascending order. CEB one curve per panel; EPB three ($\rho \in \{1, 6, 12\}$).